Master's Thesis

Variance-Based Analysis of Predictive Multiplicity in Rashomon Sets

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Abstract

Modern machine learning pipelines often produce many models with near-identical predictive performance, yet these models can disagree substantially on individual predictions—a phenomenon known as predictive multiplicity. Existing approaches quantify this disagreement at the aggregate level but do not reveal where in the feature space it concentrates or which modeling choices are associated with it.

This thesis introduces a spatial analysis framework for predictive multiplicity. We compute observation-wise predictive variance across finite top-K Rashomon approximations selected by validation Brier score and analyze its structure using spatial statistics, including global Moran's I and Local Indicators of Spatial Association (LISA). This enables the detection, localization, and statistical assessment of multiplicity hotspots—regions in feature space where predictions are particularly model-dependent. We further decompose predictive variance into between-family and within-family hyperparameter components and analyze how these descriptive sources of variation differ across the feature space.

Experiments on three benchmark datasets with five model families show that, under the tested benchmark settings and the prediction-permutation null, predictive multiplicity is often spatially structured. Model family accounts for a substantial share of observed disagreement, with stronger hotspot-specific contributions on COMPAS and Adult. These patterns remain visible under changes in Rashomon set size, neighborhood definition, probability calibration, and graph construction. Synthetic experiments with known ambiguous regions validate the pipeline's ability to recover ground-truth instability. High-multiplicity regions can additionally be characterized by compact, interpretable rules in the original feature space.

The results suggest that predictive multiplicity is not merely a global property of model classes, but can also appear as a localized phenomenon affecting specific regions of the feature space. The proposed framework provides a diagnostic for identifying predictions that are especially sensitive to the choice among near-optimal models, with implications for the reliability and governance of automated decision systems.

Contents

1 Introduction

Supervised machine learning (ML) models are increasingly used to support decisions in domains such as healthcare, finance, and criminal justice. Their appeal lies in strong predictive performance, yet high average performance does not guarantee that individual predictions are stable or trustworthy. In modern ML pipelines, many distinct models often achieve nearly identical performance on the same task, while still producing substantially different predictions for particular observations. Classical evaluation metrics such as accuracy, AUC (Area Under the Curve), or Brier score summarize overall model quality, but they provide no insight into whether a specific prediction is robust across equally acceptable models.

This phenomenon is known as *predictive multiplicity*: multiple models that are essentially indistinguishable in overall performance can still disagree in meaningful ways at the level of individual decisions. As a result, a prediction may reflect not only the data, but also an arbitrary modeling choice within a broader set of near-optimal alternatives. In high-stakes settings, this raises a natural question: are all predictions from a well-performing model equally trustworthy, or are some individuals especially exposed to model-dependent instability?

In this thesis, we study predictive multiplicity through a spatial lens. Since each observation can be represented as a point in feature space, disagreement can be studied not only observation by observation, but also in terms of whether nearby observations share similar levels of instability. We therefore ask not only *how much* near-optimal models disagree on average, but also whether this disagreement is concentrated in specific regions of the feature space, whether such regions are stable, and which modeling choices are associated with them. To answer these questions, we propose a framework that combines observation-wise predictive variance with tools from spatial statistics to detect, localize, and interpret regions of concentrated instability.

1.1 Motivation

Consider an automated risk assessment system used to support decisions about individual cases. For one person, the deployed model predicts a risk score of 0.29 and therefore classifies the case as low risk. During an audit, however, the analyst does not only inspect the deployed model, but also other models that achieve nearly the same validation performance. For the same person, these alternative models produce risk scores ranging from 0.22 to 0.54. The final classification may therefore depend on which near-optimal model happened to be selected.

This example illustrates why aggregate performance alone is not enough in high-stakes settings. A model may perform well on average, while some individuals remain highly sensitive to model choice. For such cases, the relevant question is not only whether the deployed model is accurate overall, but whether the individual prediction is stable across equally acceptable alternatives.

The same audit can also be extended beyond a single individual. If many nearby cases in feature space show similarly high prediction instability, then the problem is not an isolated exception but a localized region of model-dependent decisions. Such a region may correspond to a subpopulation for which predictions are systematically less robust. This raises

the practical need to identify where these regions occur and to understand whether they are mainly associated with broad model-family choices or with hyperparameter variation within families.
This thesis is motivated by this audit perspective: predictive multiplicity should not only be measured globally, but localized, tested for spatial structure, and related back to the modeling choices that generate it.

1.2 Problem Statement and Research Questions

Models with nearly identical overall performance can still differ substantially in their predictions for individual observations. This can happen in ordinary machine learning workflows: different model types may be tried, hyperparameters may be tuned in different ways, and random seeds may lead to slightly different fitted models. Standard evaluation metrics summarize how well a model performs on average, but they do not show whether a particular prediction would remain similar if another equally good model had been chosen.
This is problematic in automated decision systems because decisions are made for individual cases. A prediction can appear reliable when only the deployed model is considered, even though other models with similar validation performance would give a different score.
This thesis therefore studies predictive multiplicity at the level of individual observations. It focuses on where unstable predictions occur in feature space, whether they appear as isolated cases or as local regions, and which modeling choices are associated with them.
The thesis addresses the following questions:

1. **Where do similarly well-performing models disagree?** How much do predicted probabilities vary across near-optimal models for individual observations, and which observations are most affected?

2. **Does disagreement form local regions in feature space?** Are observations with high prediction instability scattered across the test set, or do nearby observations tend to be unstable together?

3. **How stable are these regions under changes in the analysis?** When high-instability regions are detected, do they still appear when the data split changes, when a different number of near-optimal models is selected, when the neighborhood definition in feature space is changed, or when predicted probabilities are calibrated?

4. **Which modeling choices are associated with the instability?** Is prediction instability mainly related to broad choices between model types, such as logistic regression, random forests, or neural networks, or to tuning choices within a model type, such as regularization strength, tree depth, or learning rate?

5. **Can the detected regions be validated and interpreted?** Does the approach recover known ambiguous regions in synthetic data, and can high-instability regions in real datasets be summarized by simple rules in the original input features?

1.3 Thesis Structure

The remainder of the thesis is structured as follows. Chapter 2 introduces the main concepts needed for the thesis and reviews related work on predictive multiplicity, Rashomon sets, and spatial statistics. Chapter 3 presents the framework and experimental design. It defines how the near-optimal model set is constructed, how prediction instability is measured, how feature-space hotspots are detected, and how the experiments, robustness checks, and attribution analyses are implemented.

Chapter 4 presents the empirical results. It first reports how much the selected models disagree on the real-world benchmark datasets, then analyzes whether this disagreement forms local regions, how stable these regions are, and which modeling choices are associated with them. The chapter also includes synthetic experiments, where the unstable regions are known in advance, and rule-based descriptions of selected high-instability regions. Chapter 5 discusses the main findings, limitations, and practical implications. Chapter 6 summarizes the thesis and outlines possible directions for future work.

2 Background and Related Work

Predictive multiplicity lies at the intersection of three lines of work: the study of Rashomon sets, the design of metrics that quantify disagreement among near-optimal models, and the broader question of what such disagreement implies for model selection, fairness, and reliability. The central observation is that a single deployed model is often only one choice among several similarly well-performing alternatives. Even when a machine-learning workflow ultimately returns one final model, other choices of model family, hyperparameters, or random seed may lead to models with nearly identical predictive performance but different outputs for the same individual. This chapter reviews the literature most directly connected to that observation and uses it to motivate the perspective taken in this thesis: predictive multiplicity should be studied not only as a global property of a dataset or model class, but also as a localized phenomenon in feature space.

2.1 Rashomon Sets and Predictive Multiplicity

The modern formulation of multiplicity is rooted in the *Rashomon effect*, introduced by Breiman (2001). The Rashomon effect describes the situation in which many distinct models achieve similarly good predictive performance, while relying on different functional forms, variables, or relationships in the data. In other words, there may be no single uniquely best explanation of the data-generating relationship. This perspective is also used by Fisher et al. (2019), who argue that conclusions drawn from one fitted model can be incomplete when other well-performing models rely on different variables or relationships.

Formally, let $\mathcal{F}$ be a hypothesis class, let $L(f)$ be the loss of a model $f \in \mathcal{F}$, and let $L^* = \min_{f\in\mathcal{F}} L(f)$ be the best loss in this class. The corresponding *Rashomon set* is

$$\mathcal{R}_\varepsilon = \big\{f \in \mathcal{F} : L(f) \leq L^* + \varepsilon\big\}.$$

It contains all models whose loss is within a tolerance ε of the best model. This shifts attention from a single selected model to a set of near-optimal alternatives.

Empirically, this thesis approximates Rashomon sets by finite top-K candidate pools ranked by validation Brier score; conclusions therefore concern this operational approximation rather than the full tolerance-defined Rashomon set.

Recent Rashomon-set research has emphasized that the size and structure of $\mathcal{R}_\varepsilon$ can have substantial practical consequences. Semenova et al. (2022) formalize the *Rashomon ratio*, i.e., the volume of $\mathcal{R}_\varepsilon$ relative to the hypothesis space, and argue that large Rashomon sets help explain why simpler yet accurate models may exist. Rudin et al. (2024) broaden this perspective further: when many good models are available, model development should not be viewed as a search for a uniquely correct solution, but as a search over a set of viable models that may differ in interpretability, fairness, robustness, and other secondary properties.

Within this broader setting, Marx et al. (2020) introduce *predictive multiplicity* as the phenomenon that near-optimal models can assign conflicting predictions to the same observation. Black et al. (2022) generalize this concern under the broader notion of *model multiplicity*, highlighting that arbitrariness in model choice can create both opportunities

and risks: multiplicity may enable deliberate optimization for fairness or interpretability, but it may also make individual outcomes depend on choices that are invisible to affected stakeholders.
Predictive multiplicity should also be distinguished from classical notions of uncertainty. Aleatoric uncertainty reflects inherent noise in the data, while epistemic uncertainty captures limited knowledge about the data-generating process. In contrast, predictive multiplicity arises from the existence of multiple equally accurate models that produce different predictions for the same observation. The resulting disagreement is therefore not due only to noise or lack of information, but to the structure of the model space itself.
This thesis studies precisely that predictive facet of multiplicity: how much near-optimal models disagree on individual observations, where that disagreement is concentrated, and which modeling choices are associated with it.

2.2 Metrics for Predictive Multiplicity

The first generation of predictive-multiplicity metrics operates in *label space*. Marx et al. (2020) propose *ambiguity*, which measures how often a sample can receive conflicting labels within the Rashomon set, and *discrepancy*, which quantifies the maximal extent of such disagreement. These measures made predictive multiplicity operational, but they reduce model outputs to binary decisions based on a fixed threshold (e.g., 0.5), thereby ignoring differences in predicted probabilities when models agree on the final label.
A second line of work extends multiplicity to *probability space*. Watson-Daniels et al. (2023) develop a framework for predictive multiplicity in probabilistic classification, introducing score-based measures that capture the variation of risk estimates across near-optimal models rather than only label flips. Their analysis also connects multiplicity to dataset characteristics such as separability, outliers, and majority–minority structure. This is especially relevant for this thesis because the main response variable is not hard disagreement alone, but the full spread of predicted probabilities across the Rashomon set.
Closely related, Hsu and Calmon (2022) propose *Rashomon Capacity*, an information-theoretic measure of predictive multiplicity in probabilistic classification. The measure is defined at the level of an individual observation and captures how much the possible predicted scores for that observation vary across models in the Rashomon set. Hsu et al. (2024) extend this line in the context of gradient boosting, introducing *RashomonGB* as a practical method for inspecting multiplicity in boosted models and discussing score-based notions such as the *Viable Prediction Range* for individual observations.
At the per-observation level, one particularly useful hard-disagreement measure is the *conflict ratio*, which records the fraction of models on the minority side of a decision threshold. Recent work by Frohnapfel et al. (2026) argues that such individual-level multiplicity measures are not only methodologically useful but also policy-relevant, because they can help operationalize per-person performance reporting requirements in high-stakes settings under the EU AI Act.
Taken together, the literature now offers several ways to quantify multiplicity globally and per instance, in both label space and probability space. What remains much less developed is how to move from pointwise disagreement scores to an analysis of whether

disagreement forms coherent *regions* in feature space.

2.3 Localizing Instability in Feature Space

The metrics discussed above quantify predictive multiplicity either at the dataset level or for individual observations. A remaining question is whether observations with high prediction disagreement are isolated cases, or whether they form coherent regions in feature space. This thesis approaches that question using tools from spatial statistics.

Moran (1950) introduced Moran's I as a measure of global spatial autocorrelation. In this thesis, it is used to test whether observations with similar levels of prediction instability tend to be close to each other in feature space. Anselin (1995) developed *Local Indicators of Spatial Association* (LISA), which identify where such local clustering occurs. In particular, LISA can identify *High–High* regions, where observations with high values are surrounded by neighbors that also have high values, and *Low–Low* regions, where low values cluster together.

In their original setting, these tools operate on geographic data with an explicit spatial adjacency structure. In this thesis, the same logic is transferred to *feature space*: observations become nodes in a k-nearest-neighbor graph, and observation-wise multiplicity scores become the variable whose spatial organization is tested. Conceptually, this turns predictive multiplicity from a collection of isolated pointwise values into a structured signal that can be analyzed for clustering, significance, and local hotspot formation.

This step is motivated by observations already present in the literature on predictive multiplicity. Watson-Daniels et al. (2023) show that prediction disagreement across near-optimal probabilistic classifiers is associated with dataset characteristics such as outliers and separability. This suggests that predictive multiplicity may not be distributed uniformly across the input space. Likewise, per-instance feasible-range methods such as those discussed by Hsu et al. (2024) show that the range of possible predictions can vary substantially from one observation to another. These works therefore motivate an observation-level view of predictive multiplicity. However, they do not directly analyze whether observations with high disagreement form statistically significant, spatially contiguous regions in feature space.

Recent work by Haghighat et al. (2026) also considers local regions around test points, but with a different goal: their local patching and reconciliation methods aim to reduce disagreement among Rashomon-set models. In contrast, this thesis treats local structure as the object of diagnosis, testing whether predictive variance is spatially autocorrelated in feature space and whether high-multiplicity regions recur as graph-based hotspots.

That question matters for interpretation. Once disagreement is shown to cluster, the relevant object is no longer only an unstable individual prediction, but an unstable *subpopulation*. This shifts the focus from isolated observations to coherent regions of the feature space, enabling a more structured analysis of where and how predictive multiplicity arises.

2.4 Drivers and Consequences of Predictive Multiplicity

After asking how predictive multiplicity can be measured and localized, a further question is why it arises and why it matters. One source is model selection itself. If several models perform similarly well, then choosing one final model is not only a technical step, but can affect individual predictions. Black et al. (2022) and Rudin et al. (2024) emphasize this point: when many good models are available, model choice should be seen as a substantive design decision, because different acceptable models may differ in other properties such as interpretability, fairness, or robustness.

More recently, attention has turned to concrete modeling choices that generate multiplicity within a model family. Most directly relevant for this thesis is Cavus et al. (2025), who show across benchmark datasets that hyperparameter tuning can materially increase prediction discrepancy even when performance improves. Their results suggest that multiplicity is not driven only by broad differences between families such as linear models versus tree ensembles, but also by movement through hyperparameter space within a fixed family.

A related line of work studies whether post-hoc calibration can reduce predictive multiplicity. Cavus (2026) investigates calibration methods such as Platt scaling, isotonic regression, and temperature scaling in classification settings and finds that calibration can reduce aggregate multiplicity measures across near-optimal models. This perspective treats calibration as a possible mitigation strategy. In contrast, the calibration analysis in this thesis is used as a robustness diagnostic: it asks whether the spatial clustering of predictive variance persists after the probability scale of each model has been adjusted.

Recent work also highlights that prediction variance across models can lead to arbitrary decisions for individual observations (Cooper et al., 2024).

The implications of such disagreement have also been studied from a fairness and governance perspective. Sokol et al. (2024) formulate *cross-model fairness*, emphasizing that when different equally performing models treat the same person differently, fairness concerns arise not only from the deployed predictor itself but also from the unresolved choice among competing predictors. Similarly, Frohnapfel et al. (2026) connect predictive multiplicity to legal and reporting obligations in high-risk decision support, arguing that disagreement across near-optimal models can be interpreted as a form of person-level arbitrariness that should be made visible to deployers.

These works clarify why multiplicity matters and which modeling choices may create it. However, they mostly operate at the level of whole datasets, whole model classes, or individual cases taken one at a time. They do not yet ask whether family effects and hyperparameter effects become especially strong in *specific parts* of feature space.

2.5 Positioning of This Thesis

The existing literature provides concepts and metrics for studying predictive multiplicity, but it mostly treats disagreement either as a dataset-level quantity or as a property of individual observations. Less attention has been given to whether these observation-level differences form coherent regions in feature space.

This thesis addresses this gap by treating prediction instability as a spatial signal. It uses observation-wise predictive variance to identify where near-optimal models disagree, and then analyzes whether high-disagreement observations cluster locally. The goal is therefore not only to measure how much predictive multiplicity exists, but also to locate where it occurs, assess how stable these regions are, and examine which modeling choices are associated with them.

In this way, the thesis connects predictive multiplicity with a regional view of model reliability: some parts of the feature space may be more sensitive to model choice than others, even when all considered models perform similarly well overall.

3 Framework and Experimental Design

This chapter defines the analysis framework and its experimental implementation. The main goal is to make predictive multiplicity observable at the level of individual test observations and then to test whether this instability is spatially organized in feature space. The chapter first describes how near-optimal models are collected and how observation-wise disagreement is measured. It then defines the feature-space graph, Moran's I, LISA hotspot detection, the permutation null model, and the additional robustness, validation, and attribution analyses used in the empirical study.

3.1 Overview of the Analysis Pipeline

The analysis starts from a supervised binary classification dataset and a finite pool of trained candidate models. Candidate models are ranked by validation Brier score, and the main Rashomon approximation is defined as the top-K models in this ranking. For each test observation, the selected models produce a vector of predicted probabilities. The primary multiplicity measure is the variance of these predicted probabilities across the selected models.

The resulting vector of observation-wise predictive variance values is then analyzed as a spatial signal on a graph. The graph is constructed in the preprocessed feature space of the test set: observations are nodes, and edges connect nearby observations according to a symmetrized k-nearest-neighbor (kNN) relation. Global Moran's I is used to test whether high-variance observations tend to lie near other high-variance observations. Local Moran statistics, or Local Indicators of Spatial Association (LISA), are then used to identify local High–High (HH) regions. These HH regions are interpreted as multiplicity hotspots: observations with above-average predictive variance surrounded by neighbors that also have above-average predictive variance.

Finally, the observed spatial structure is compared with a prediction-permutation null model. This null preserves each model's marginal distribution of predictions but destroys the alignment between predictions and feature-space location. The empirical analyses then ask whether the detected hotspot structure is stable across runs, robust to analytical choices, recoverable in synthetic data, and attributable to model-family or hyperparameter choices.

3.2 Data, Model Pool, and Rashomon Approximation

Problem setting and notation. Let (X, y) denote a supervised learning dataset with N observations, where $X \in \mathbb{R}^{N\times d}$ is the feature matrix and $y \in \{0,1\}^N$ is a binary target variable. Let $\mathcal{F}$ denote a finite pool of trained probabilistic classifiers, potentially spanning multiple model families and hyperparameter configurations. Each model $f \in \mathcal{F}$ outputs predicted probabilities $\hat{p}_{fi} \in [0,1]$ for observations $i = 1,\dots,N$.

Datasets. Experiments are conducted on three real-world benchmark datasets from different application domains. All three tasks are binary classification problems and contain both numerical and categorical input features.

- **COMPAS**: The ProPublica COMPAS recidivism dataset (Angwin et al., 2016, Larson, 2016). It contains defendant-level information from Broward County, Florida, such as age, prior offence count, sex, race, and charge degree. The target indicates whether the individual reoffended within two years.
- **German Credit**: The `credit-g` dataset from OpenML (Hofmann, 1994). It contains financial and demographic information about credit applicants. The target indicates good versus bad credit risk.
- **Adult**: The Adult income dataset from OpenML (Kohavi and Becker, 1996). It contains demographic and employment-related information. The target indicates whether annual income exceeds $50,000.

The datasets differ in size, domain, and feature composition. Additional dataset characteristics, including split sizes, target prevalence, and transformed feature dimensionality, are reported in Appendix B. These differences provide a basic check of whether the proposed spatial multiplicity analysis behaves similarly across different tabular classification settings.

Data splitting. Each dataset is split into training (60%), validation (20%), and test (20%) sets using stratified sampling. The training set is used to fit all candidate models. The validation set is used to rank candidates by Brier score and to construct the finite top-K model set. The test set is held out until the final analysis step and is used only to compute predictive variance and spatial statistics.

The full experiment is repeated with 10 random seeds. Two splitting modes are used. For the main analyses, each seed creates a new stratified train–validation–test split; this reflects variability in the full pipeline, including model fitting, validation-based selection, test-set prediction, and spatial analysis. For the cross-seed stability analysis, the test set is drawn once and kept fixed across seeds while the training and validation parts are re-split. This fixed-test-set protocol makes pointwise comparisons such as HH selection frequency and Jaccard overlap meaningful, because the same test observations are compared across runs.

Model families, training, and preprocessing. All models are implemented in Python using `scikit-learn` (Pedregosa et al., 2011). The candidate pool contains five model families: logistic regression with elastic-net regularization, k-nearest neighbors (kNN), random forest, gradient boosting, and multilayer perceptron (MLP). These families cover a linear model, an instance-based method, two tree-based ensemble methods, and a neural-network model.

For each model family and run, 50 hyperparameter configurations are sampled from predefined finite search spaces. The full search spaces are reported in Appendix A. Configurations are sampled using deterministic seeds derived from the run. If duplicate configurations occur for families with smaller discrete search spaces, configurations are resampled until 50 unique candidates are obtained. No cross-validation is used inside a run; instead, the full pipeline is repeated across 10 outer train–validation–test splits.

Preprocessing is applied uniformly across all model families via a column transformer fitted on the training set only. Numeric features are imputed with the training-set median and standardized to zero mean and unit variance. Categorical features are imputed with the training-set mode and one-hot encoded; unknown categories at validation or test time are mapped to a zero vector. Columns not identified as numeric or categorical are dropped. The fitted transformer is applied to validation and test data without refitting, thereby avoiding data leakage. The same transformed feature space is used for model training and for constructing the feature-space graph.

Rank-based Rashomon approximation. The classical Rashomon set is usually defined through a loss tolerance around the best model (Breiman, 2001, Marx et al., 2020, Semenova et al., 2022). In this thesis, we use a finite, rank-based approximation instead. The empirical model set is therefore not defined by a fixed absolute or relative performance tolerance, but by selecting a fixed number of near-optimal models from the trained candidate pool.

Each model $f \in \mathcal{F}$ is evaluated on the validation set using a loss $L(f)$. In this thesis, L is the Brier score, because the subsequent multiplicity analysis is based on predicted probabilities rather than only on hard class labels. Let $f_{(1)}, \ldots, f_{(|\mathcal{F}|)}$ denote the models sorted by increasing validation loss, so that

$$L(f_{(1)}) \le L(f_{(2)}) \le \cdots \le L(f_{(|\mathcal{F}|)}).$$

The main empirical Rashomon approximation is

$$\mathcal{R}_K = \{f_{(1)}, \ldots, f_{(K)}\}.$$

Unless otherwise stated, $K = 25$ and $\mathcal{R}_K$ refers to the global top-K set selected from the full candidate pool across all families.

The reason for using a fixed K rather than a fixed ε-based performance threshold is comparability. A fixed tolerance can lead to very different numbers of selected models across datasets, splits, and model families. This would make observation-wise variance estimates, spatial hotspot analyses, and comparisons across runs harder to interpret. Fixing K ensures that each run is based on the same number of near-optimal models while retaining the central idea of predictive multiplicity: studying disagreement among models with very similar validation performance.

To audit how close the retained models are in validation performance, we report the validation Brier-score gap within the selected global top-K set. For each run, this gap is computed relative to the best retained model,

$$\Delta L_m = L(f_m) - \min_{f \in \mathcal{R}_K} L(f), \qquad f_m \in \mathcal{R}_K.$$

Table 1 summarizes the maximum, median, and 95th-percentile gap across the selected top-$K = 25$ models. This diagnostic makes the term “near-optimal” explicit for the finite rank-based approximation used in the experiments.

The gaps are small in absolute Brier-score units across all three datasets. German Credit has the widest selected top-K range, but even there the average maximum gap remains below 0.01. This supports interpreting the selected models as a near-optimal finite candidate

Table 1: Validation Brier-score gaps within the global top-$K = 25$ Rashomon approximation. For each run, gaps are computed relative to the best retained model. Values are mean $\pm$ standard deviation across runs.

Dataset	Maximum gap	Median gap	95th-percentile gap
COMPAS	0.0021 ± 0.0007	0.0013 ± 0.0005	0.0020 ± 0.0006
German Credit	0.0076 ± 0.0025	0.0058 ± 0.0020	0.0073 ± 0.0027
Adult	0.0039 ± 0.0006	0.0018 ± 0.0003	0.0037 ± 0.0006

set, while making clear that near-optimality is defined operationally by the rank-based top-K construction.

For analyses that compare model families or study variation within one family, we also use family-specific top-K sets. If $\mathcal{F}_g \subset \mathcal{F}$ denotes the candidate models belonging to family g, then $\mathcal{R}_K^{(g)}$ denotes the top-K models within that family according to validation Brier score. This construction retains enough models from each family for the family-wise and within-family attribution analyses.

3.3 Measuring Observation-Wise Predictive Multiplicity

For each run, all models in the selected Rashomon approximation are applied to the held-out test set. This yields a prediction matrix

$$P \in \mathbb{R}^{K \times N},$$

where K is the number of selected models and N is the number of test observations. The entry $\hat{p}_{mi}$ denotes the predicted probability assigned to test observation i by model $m \in \mathcal{R}_K$.

The primary observation-wise multiplicity metric is predictive variance. Let $M = |\mathcal{R}_K|$ and let

$$\bar{p}_i = \frac{1}{M} \sum_{m=1}^{M} \hat{p}_{mi}$$

denote the mean predicted probability for observation i across the retained models. The observation-wise predictive variance is computed as the population variance over the finite selected model set:

$$v_i = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{p}_{mi} - \bar{p}_i\right)^2, \quad i = 1, \dots, N.$$

Thus, the denominator is M rather than $M - 1$. This convention matches the empirical pipeline, where the selected top-K model set is treated as the finite Rashomon approximation under study rather than as a random sample from a larger population.

High values of v_i indicate that near-optimal models assign substantially different probabilities to observation i, while low values indicate agreement. This follows the score-based view of predictive multiplicity, where disagreement is measured at the level of predicted scores or probabilities rather than only after thresholding into class labels (Watson-Daniels et al., 2023, Hsu and Calmon, 2022).

An alternative score-based measure is *Rashomon Capacity* (Hsu and Calmon, 2022). Like predictive variance, it is defined at the level of an individual observation and captures variation in predicted scores across near-optimal models. It uses an information-theoretic formulation, whereas predictive variance uses the empirical spread of predicted probabilities in the finite model set. Predictive variance is used here because it is directly computable from $\mathcal{R}_K$, easy to interpret as probability spread, and can be decomposed naturally in the later family- and hyperparameter-level analyses.

In addition to pointwise predictive variance, the results report aggregate multiplicity summaries. Mean variance averages v_i across test observations. Ambiguity is the mean prediction range across Rashomon models, $\max_m \hat{p}_{mi} - \min_m \hat{p}_{mi}$. The disagreement rate is the fraction of test observations for which model predictions differ by more than $\delta = 0.05$. Discrepancy is the maximum, over model pairs, of the mean absolute prediction difference. A supplementary decision-level analysis based on the conflict ratio is reported in Appendix G; it is not part of the main pipeline.

3.4 Feature-Space Graph and Spatial Hotspot Detection

To study whether predictive variance is locally concentrated, the test observations are represented as a graph in feature space. Each node is one test observation, and edges connect observations that are similar according to the preprocessed input features. Throughout this thesis, “spatial” therefore refers to neighborhoods in feature space, not to geographic space.

Graph construction. For each dataset and run, the graph is constructed on the held-out test set. Euclidean distances are computed between preprocessed test-set feature vectors. For each observation, its k nearest neighbors are identified, yielding an initial directed kNN graph. Because nearest-neighbor relations are not necessarily symmetric, the graph is symmetrized: an undirected edge is placed between observations i and j if either observation is among the k nearest neighbors of the other.

Let

$$A \in \{0,1\}^{N\times N}$$

denote the resulting symmetric adjacency matrix, where $A_{ij} = 1$ means that observations i and j are connected. The adjacency matrix is row-standardized to obtain the spatial weight matrix W:

$$W_{ij} = \frac{A_{ij}}{\sum_{j'=1}^{N} A_{ij'}}, \quad i = 1, \ldots, N.$$

After this normalization, each row of W sums to one. Therefore, for an observation-level quantity $\mathbf{v} = (v_1, \ldots, v_N)^\top$, the spatial lag

$$[W\mathbf{v}]_i = \sum_{j=1}^{N} W_{ij} v_j$$

is the average value of $\mathbf{v}$ among the neighbors of observation i. If v_i is predictive variance, then $[W\mathbf{v}]_i$ is the average predictive variance in the local neighborhood of observation i.

The baseline graph therefore encodes one specific operational notion of local similarity: Euclidean distance in the preprocessed feature representation. This choice is natural because it uses the same transformed feature space as the predictive models, but it is not unique. In particular, distances in one-hot encoded tabular data can depend on the chosen preprocessing and scaling. For this reason, the robustness analysis also compares the baseline graph with PCA-reduced and cosine-distance alternatives (Section 4.4.4).
In the experiments, the reported k values refer to the initial directed nearest-neighbor query. After symmetrization, individual nodes may have more than k neighbors. The default values are $k = 30$ for COMPAS and German Credit and $k = 60$ for Adult, where the larger value helps avoid disconnected graph components.

Global Moran's I. Global Moran's I is a classical measure of spatial autocorrelation (Moran, 1950). It measures whether similar values tend to occur near each other in a given neighborhood graph. Here it tests whether high-variance observations tend to be close to other high-variance observations, and whether low-variance observations tend to be close to other low-variance observations.
Let $\mathbf{v} = (v_1, \ldots, v_N)^\top$ denote the predictive variance values on the test set, and let $\bar{v}$ be their mean. Because W is row-standardized, Moran's I can be written as

$$I = \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} W_{ij}(v_i - \bar{v})(v_j - \bar{v})}{\sum_{i=1}^{N}(v_i - \bar{v})^2}.$$

If observations with above-average variance tend to have neighbors that are also above average, the numerator becomes positive and Moran's I is positive. Values close to zero indicate little systematic organization with respect to the graph. Negative values would indicate that high values tend to be surrounded by low values, or vice versa.

Local Moran statistics and LISA. Global Moran's I summarizes the overall degree of spatial clustering, but it does not show where clustering occurs. For this reason, we also use Local Indicators of Spatial Association (LISA), introduced by Anselin (1995). LISA decomposes spatial association into observation-level statistics and identifies observations that contribute to local clusters.
In the PySAL implementation, the input variable is standardized before local Moran statistics are computed. Let

$$z_i = \frac{v_i - \bar{v}}{s_v}$$

denote the standardized predictive variance of observation i, where s_v is the standard deviation of predictive variance across test observations. The spatial lag of the standardized values is

$$[W\mathbf{z}]_i = \sum_{j=1}^{N} W_{ij} z_j.$$

The local Moran statistic used by PySAL is proportional to

$$z_i [W\mathbf{z}]_i,$$

up to a constant scaling factor. It is large and positive when observation i and its neighbors are on the same side of the mean, for example when both have above-average predictive variance.
For each observation, a permutation test compares the observed local statistic with a reference distribution obtained by randomly rearranging neighboring values. This produces one local p-value per observation. Because many local tests are performed at once, the Benjamini–Hochberg procedure is applied to control the false discovery rate (Benjamini and Hochberg, 1995). Observations that remain significant after this correction are assigned to one of four local spatial categories:

- **High–High (HH):** $z_i > 0$ and $[W\mathbf{z}]_i > 0$. The observation has above-average predictive variance and is surrounded by neighbors with above-average predictive variance. These observations are interpreted as multiplicity hotspots.
- **Low–Low (LL):** $z_i < 0$ and $[W\mathbf{z}]_i < 0$. The observation has below-average predictive variance and is surrounded by neighbors with below-average predictive variance. These observations mark locally stable regions.
- **High–Low (HL):** $z_i > 0$ and $[W\mathbf{z}]_i < 0$. The observation has above-average predictive variance but its neighbors have below-average predictive variance. This corresponds to a high-variance spatial outlier.
- **Low–High (LH):** $z_i < 0$ and $[W\mathbf{z}]_i > 0$. The observation has below-average predictive variance but its neighbors have above-average predictive variance. This corresponds to a low-variance observation inside a high-variance neighborhood.

Observations that are not significant after correction are labeled non-significant. An *HH region* refers to the set of High–High observations. An *HH hotspot* refers to a local region of elevated predictive variance.

Connected HH components. The LISA classification identifies individual HH observations. To obtain larger coherent regions, the symmetrized kNN graph is restricted to HH observations only and connected components are extracted. A connected component is a set of HH observations connected by paths in the graph. Each component is interpreted as a contiguous high-multiplicity region in feature space. Very small components may be discarded to avoid treating isolated fragments as substantive hotspot regions.
For the standard spatial-statistics computations, PySAL's value-permutation tests are used: global Moran's I is assessed with 999 permutations, and Local Moran (LISA) also uses 999 permutations per observation. These permutations assess spatial autocorrelation of the already-computed predictive-variance field on the fixed feature-space graph. For LISA, the resulting local p-values are corrected with the Benjamini–Hochberg procedure at $\alpha = 0.05$ before assigning HH and LL labels. This PySAL value-permutation procedure is distinct from the prediction-matrix permutation null introduced in the next section.

3.5 Permutation Null Model

In addition to the standard PySAL value-permutation tests used for Moran's I and LISA, we use a separate prediction-matrix permutation null. This second null is designed to test whether the spatial clustering of predictive variance depends on the alignment between model predictions and feature-space locations.

The null operates on the prediction matrix $P \in \mathbb{R}^{K \times N}$. For each model m independently, the vector $(\hat{p}_{m1}, \ldots, \hat{p}_{mN})$ is randomly permuted across observations. This within-model permutation preserves each model's marginal distribution of predictions, but destroys systematic alignment between predictions and the feature-space graph. Predictive variance is then recomputed from the permuted prediction matrix, and global Moran's I is recalculated. Any spatial clustering in the recomputed variance field is therefore interpreted relative to a null in which predictions retain their model-specific marginal distributions but are no longer attached to their original feature-space locations.

For each of R permutations, observation-wise predictive variance and global Moran's I are recomputed. The empirical one-sided p-value is

$$p = \frac{1 + \#\{\text{null} \geq \text{observed}\}}{R + 1}.$$

In the experiments, $R = 100$ null permutations are used per run. A run is treated as significant when the empirical p-value is below 0.05. Because $R = 100$, the smallest attainable empirical p-value is $1/(R+1) = 1/101 \approx 0.0099$. Reported values of "$p < 0.01$" should therefore be interpreted at this resolution limit.

3.6 Robustness and Validation Design

Several additional analyses test whether the observed spatial structure is robust, methodologically stable, and recoverable under controlled conditions.

Sensitivity to Rashomon set size. The default global Rashomon approximation uses $K = 25$. Sensitivity to this choice is examined by varying $K \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ using global top-K selection. For each K, predictive variance, Moran's I, and HH counts are recomputed.

Sensitivity to neighborhood size. The default graph uses dataset-specific k values. To assess whether conclusions depend on this choice, the neighborhood size is varied and the variance-based spatial metrics are recomputed. Mean predictive variance itself is independent of k, but Moran's I and HH counts depend on the graph definition.

Calibration robustness. To test whether the spatial hotspot structure is mainly driven by probability miscalibration, we apply Platt scaling and isotonic regression to the predictions of each model in the global Rashomon set. For each model, a logistic regression is fitted on validation-set predicted probabilities and validation labels, and the resulting mapping is applied to that model's test-set predictions. The predictive-variance and spatial pipeline is then rerun on both uncalibrated and calibrated test predictions.

We record changes in mean variance, Moran's I, HH/LL masks, connected components, and the Jaccard similarity between hotspot masks before and after calibration. Calibration is used here as a post-selection robustness check rather than as an independent performance evaluation. Because the validation set is used both for selecting the top-K model set and for fitting the calibration mappings, calibrated metrics are interpreted descriptively. Synthetic datasets are excluded from this analysis because they follow a separate data-generating design.

Alternative graph constructions. To verify that the spatial results are not artifacts of Euclidean distance on the full one-hot encoded feature space, the baseline graph is compared with two alternatives: (i) a kNN graph in PCA-reduced space retaining 5 components, and (ii) a kNN graph using cosine distance on ℓ_2-normalized features. All other pipeline parameters are kept fixed, and Moran's I and HH counts are compared across graph definitions.

Synthetic validation. Two synthetic data designs are used to validate whether LISA-based HH regions recover known ambiguous regions. The single-island design contains two well-separated stable Gaussian blobs and one ambiguous island, with points drawn uniformly in a disk and weak XOR-like conditional probability $P(y = 1 \mid x) = 0.5 \pm \delta$. The three-islands design contains two stable blobs, three ambiguous islands, and a small number of isolated outliers with known region labels. The same training pipeline is applied: a 60/20/20 split, the same five model families, and top-K selection by validation Brier score. Recovery of the ground-truth ambiguous regions by LISA HH classification is evaluated using precision, recall, and Jaccard overlap. The prediction-permutation null is used to verify that observed Moran's I is significant.

3.7 Attribution and Interpretation Analyses

The attribution analyses ask which modeling choices are associated with predictive multiplicity and whether hotspot regions can be described in terms of original features. All such analyses are descriptive rather than causal: they quantify how disagreement varies with model family and hyperparameter choices within the sampled candidate pool.

Global and per-family Rashomon sets. The global construction selects the top-$K = 25$ models from the full candidate pool, combining all model families. The per-family construction selects the top-$K = 25$ models separately within each family. Comparing these constructions shows whether pooling across families hides family-specific multiplicity patterns.

Family importance. Predictive variance is decomposed into a between-family component and a within-family component. Let $\mathcal{G}$ denote the set of model families and $\mathcal{R}_K^{(g)}$ the top-K model set for family $g \in \mathcal{G}$. For observation i, let $\hat{p}_{gmi}$ be the predicted probability from model $m \in \mathcal{R}_K^{(g)}$. The mean prediction of family g is

$$\bar{p}_{gi} = \frac{1}{K} \sum_{m \in \mathcal{R}_K^{(g)}} \hat{p}_{gmi},$$

and the overall mean prediction across family-specific model sets is

$$\bar{p}_i = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \bar{p}_{gi}.$$

For a set of observations S, such as all test observations, HH observations, or non-HH observations, the between-family component is

$$\mathrm{Var}_{\mathrm{between}}(S) = \frac{1}{|S|} \sum_{i \in S} \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\bar{p}_{gi} - \bar{p}_i)^2.$$

The total predictive variance across all family-specific model sets is

$$\mathrm{Var}_{\mathrm{total}}(S) = \frac{1}{|S|} \sum_{i \in S} \frac{1}{|\mathcal{G}|K} \sum_{g \in \mathcal{G}} \sum_{m \in \mathcal{R}_K^{(g)}} (\hat{p}_{gmi} - \bar{p}_i)^2.$$

Family importance is defined as

$$\mathrm{FI}(S) = \frac{\mathrm{Var}_{\mathrm{between}}(S)}{\mathrm{Var}_{\mathrm{total}}(S)}.$$

A value close to one means that most observed predictive variance is attributable to differences between model-family mean predictions. A lower value means that a larger share remains within families, for example because of hyperparameter variation. Because each family contributes the same number of models, this decomposition gives equal weight to each model family.

Within-family hyperparameter importance. After separating the between-family component, the analysis asks which grouped hyperparameter values are associated with disagreement within a given model family. For family g, let $\bar{p}_i^{(g)}$ be the mean prediction across models in $\mathcal{R}_K^{(g)}$:

$$\bar{p}_i^{(g)} = \frac{1}{K} \sum_{m \in \mathcal{R}_K^{(g)}} \hat{p}_{mi}^{(g)}.$$

For each model $m \in \mathcal{R}_K^{(g)}$, its model-level disagreement contribution is

$$V_m^{(g)} = \frac{1}{N_{\mathrm{test}}} \sum_{i=1}^{N_{\mathrm{test}}} \left(\hat{p}_{mi}^{(g)} - \bar{p}_i^{(g)}\right)^2.$$

Thus, $V_m^{(g)}$ measures how strongly model m deviates from the average prediction of its own family.

For each hyperparameter h within family g, models are grouped according to the observed value or range of that hyperparameter. Categorical hyperparameters and numeric hyperparameters with at most four distinct values are grouped by exact value. Numeric hyperparameters with more than four distinct values are discretized into three quantile bins. If the resulting groups have insufficient support, or fewer than two valid groups remain, the hyperparameter is excluded for that seed. A one-way variance decomposition of $V_m^{(g)}$ across the groups of hyperparameter h yields a between-group component $\mathrm{Var}^{(g)}_{\mathrm{between}}(h)$ and a total component $\mathrm{Var}^{(g)}_{\mathrm{total}}$. Hyperparameter importance is

$$\rho^{(g)}(h) = \frac{\mathrm{Var}^{(g)}_{\mathrm{between}}(h)}{\mathrm{Var}^{(g)}_{\mathrm{total}}}.$$

Results are aggregated over families and random seeds by mean importance, standard deviation, and the number of valid seeds.

Hotspot-specific hyperparameter shifts. To assess whether these descriptive hyperparameter associations differ inside multiplicity hotspots, the same V_m-based importance calculation is repeated on HH observations. For each hyperparameter, the hotspot-specific importance is compared with its global importance:

$$\Delta\rho(h) = \rho_{\mathrm{HH}}(h) - \rho_{\mathrm{all}}(h).$$

Positive values indicate that grouped values of the hyperparameter account for a larger share of within-family disagreement inside HH regions, while negative values indicate lower importance inside hotspots. Because HH subsets can be small, especially on German Credit, these estimates are interpreted descriptively and reported with seed coverage where relevant. An auxiliary performance-control meta-model diagnostic is reported in Appendix C.

Interpretable rule extraction. To characterize high-multiplicity regions in terms of original features, shallow decision trees with maximum depth 3 are used to extract human-readable rules. Trees are fitted on the original feature space, and positive leaf nodes are converted into conjunctive rules. The primary target label is HH membership.

Two variants are used. First, global HH rules are fitted to predict membership in the full HH set of a run. These rules describe broader subgroups enriched for HH observations but may mix several distinct hotspot regions. Second, component-level rules are fitted for connected HH components. In this variant, membership in a retained component is the positive class and all other test observations are the negative class. Component-level rules therefore aim to describe compact high-multiplicity regions rather than the full HH set at once.

Rules are evaluated using support, purity, recall, and lift. For global HH rules, purity is $P(\mathrm{HH} \mid \text{rule region})$, and lift is measured relative to the HH base rate. For component-level rules, purity is $P(\text{component} \mid \text{rule region})$, and lift is measured relative to the base rate of the corresponding component. The main rule analysis is reported for COMPAS, where the hotspot structure is clearest and retained components are sufficiently large for meaningful rule extraction.

3.8 Chapter Summary

The framework relies on four main assumptions. First, the preprocessed feature space must induce a meaningful notion of local similarity. Second, the finite candidate pool and fixed top-K selection are treated as a practical approximation of the Rashomon set. Third, the analysis is descriptive rather than causal: hotspots identify where predictive multiplicity concentrates, but do not by themselves explain the data-generating mechanisms behind that disagreement. Fourth, the framework is developed for binary probabilistic classification and would require adaptation for regression or multiclass settings.

Table 2: Overview of the main analysis goals, methods, and corresponding result sections.

Analysis goal	**Main method**	**Result section**
Measure where near-optimal models disagree	Predictive variance across $\mathcal{R}_K$	Section 4.1
Test whether disagreement is spatially clustered	Feature-space kNN graph, Moran's I, prediction-permutation null	Sections 4.1–4.2
Identify local high-multiplicity regions	LISA HH labels and connected HH components	Section 4.3
Assess stability and robustness	Fixed-test-set stability, K and k sensitivity, calibration, alternative graphs	Sections 4.3 and 4.4
Summarize associations with modeling choices	Global vs per-family sets, family importance, within-family V_m decomposition	Sections 4.5.1 and 4.5
Validate and interpret hotspots	Synthetic ground-truth designs and shallow-tree rule extraction	Sections 4.6–4.7

4 Results

This chapter presents the main empirical results. We begin with a cross-dataset overview of predictive multiplicity and its spatial structure (Section 4.1), then test the significance of the observed clustering against a permutation null and examine the stability of hotspot regions. We next assess robustness to analytical choices, including the Rashomon set size K, neighborhood size k, calibration, and graph construction. After that, we examine how model family and grouped hyperparameter values are associated with multiplicity (Section 4.5). The chapter concludes with synthetic validation and interpretable rule extraction. Unless stated otherwise, all analyses use the global Rashomon approximation (top-K with $K = 25$) and the default spatial parameters defined in Chapter 3.

4.1 Multiplicity and Spatial Metrics by Dataset

We begin with a compact cross-dataset summary of the two main questions underlying this thesis: how large predictive multiplicity is, and how strongly it is spatially structured in feature space. Table 3 reports the core summary statistics over 10 runs for COMPAS, German Credit, and Adult.

Throughout this section, we distinguish between *disagreement magnitude* and *spatial concentration*. Disagreement magnitude refers to how strongly Rashomon models disagree overall, as measured by quantities such as mean predictive variance, ambiguity, disagreement rate, and discrepancy. Spatial concentration refers to how localized this disagreement is in the feature-space graph, as measured by Moran's I, LISA HH hotspots, and connected hotspot components. Thus, a dataset can exhibit high overall disagreement without strong spatial concentration, or lower overall disagreement with clearer localized hotspot structure.

The three datasets differ clearly in both the magnitude and the spatial organization of predictive multiplicity. COMPAS exhibits the strongest spatial clustering, with the largest mean Moran's I (0.210 ± 0.081). German Credit shows weaker spatial clustering (0.088 ± 0.039), but the largest overall disagreement magnitude. Adult displays comparatively low disagreement magnitude and the weakest Moran's I (0.075 ± 0.022), though its clustering signal remains positive across runs.

The HH hotspot counts provide additional context, but should be interpreted with care. In absolute terms, Adult contains the largest number of HH observations (631.0 on average), followed by COMPAS (128.4), while German Credit contains very few HH points (5.4 on average). However, these counts are not directly comparable across datasets because both dataset size and the dataset-specific neighborhood parameter k differ. For this reason, Moran's I is the more informative cross-dataset measure of spatial concentration. Under this interpretation, COMPAS shows the clearest localized hotspot structure, whereas Adult is better characterized by broader and more diffuse regions of instability.

Table 4 reports additional aggregate multiplicity metrics that are commonly used to summarize predictive multiplicity at the dataset level.

The additional aggregate metrics confirm the same cross-dataset ordering as the predictive-variance results. German Credit exhibits the largest overall multiplicity, with the highest ambiguity, disagreement rate, and discrepancy. COMPAS lies in the middle in terms of aggregate disagreement magnitude, while Adult has the lowest values on these global

Table 3: Summary of multiplicity and spatial metrics by dataset (mean ± std over 10 runs, $K = 25$, dataset-specific default k: 30 for COMPAS and German Credit, 60 for Adult).

Dataset	Mean variance (mean ± std)	Moran's I (mean ± std)	HH count (mean ± std)	HH rate (mean ± std)
COMPAS	0.0013 ± 0.0003	0.210 ± 0.081	128.4 ± 73.6	8.9% ± 5.1%
German Credit	0.0050 ± 0.0021	0.088 ± 0.039	5.4 ± 5.4	2.7% ± 2.7%
Adult	0.0014 ± 0.0001	0.075 ± 0.022	631.0 ± 239.7	6.5% ± 2.5%

Table 4: Additional aggregate multiplicity metrics by dataset (mean ± std over 10 runs, global top-$K = 25$). Ambiguity measures the mean prediction range across Rashomon models, disagreement rate is the fraction of test observations for which predictions differ by more than $\delta = 0.05$, and discrepancy is the maximum pairwise mean absolute prediction difference between models.

Dataset	Ambiguity (mean ± std)	Disagreement rate (mean ± std)	Discrepancy (mean ± std)
COMPAS	0.125 ± 0.013	0.255 ± 0.060	0.065 ± 0.007
German Credit	0.245 ± 0.067	0.454 ± 0.156	0.138 ± 0.033
Adult	0.096 ± 0.006	0.142 ± 0.011	0.053 ± 0.007

metrics. This reinforces the main distinction of this section: disagreement magnitude and spatial concentration need not coincide. German Credit shows the strongest aggregate multiplicity, whereas COMPAS shows the clearest localized structure according to Moran's I. Adult has lower disagreement magnitude and weaker spatial clustering, but still contains a large absolute number of HH observations, reflecting its larger test set and more diffuse hotspot structure.

In the following sections, COMPAS is therefore used as the main running example for detailed spatial analyses, while German Credit and Adult are used primarily for cross-dataset comparison.

4.2 Null Hypothesis Testing

We next test whether the observed spatial clustering is stronger than expected under the prediction-matrix permutation null defined in Section 3.5. This is separate from the standard PySAL value-permutation tests used to compute Moran's I and LISA significance. In the prediction-matrix null, each model's test-set predictions are independently permuted across observations before predictive variance is recomputed. This preserves the marginal prediction distribution of each model, but detaches predictions from their original feature-space locations. The test therefore asks whether the observed Moran's I is larger than expected when prediction instability is no longer aligned with the feature-space graph.

The null test should be interpreted narrowly. It does not by itself validate the chosen feature representation, distance metric, or neighborhood graph. Those questions are addressed separately through the sensitivity analyses for the neighborhood size k and

through alternative graph constructions in Section 4.4.
Table 5 summarizes the results for predictive variance. COMPAS and Adult exhibit significant variance-based Moran's I in all runs, while German Credit is significant in 9 out of 10 runs. For COMPAS and Adult, the empirical permutation p-value reaches the minimum possible value under $R = 100$ permutations in every run, namely $p = 1/(R+1) = 1/101 \approx 0.0099$. German Credit shows the same pattern in most runs, but one run does not reach the 0.05 threshold, indicating that its spatial signal is weaker and less stable.

Table 5: Prediction-matrix permutation null summary by dataset. A run is counted as significant when the empirical permutation p-value for Moran's I is below 0.05. The null uses $R = 100$ prediction-matrix permutations per run, so the smallest attainable empirical p-value is $1/101 \approx 0.0099$. Values are computed over 10 runs.

Dataset	Runs	Significant runs	Moran's I
Adult	10	100%	0.075 ± 0.022
COMPAS	10	100%	0.210 ± 0.081
German Credit	10	90%	0.088 ± 0.039

The permutation results indicate that predictive variance is spatially clustered in feature space. After each model's test-set predictions are randomly reassigned to observations, Moran's I is consistently lower than in the observed data. Thus, high-variance observations are not randomly scattered across the test set, but tend to occur near other high-variance observations. The separation from the permutation baseline is clearest for COMPAS and Adult, while German Credit shows a weaker but still mostly significant signal.

4.3 Spatial Structure and Stability of Hotspots

Having established that predictive variance is spatially clustered, we now examine how stable the resulting HH hotspots are across runs on COMPAS. Two levels of stability must be distinguished. First, *point-level stability*: for a fixed test observation, how often is it classified as HH across random training seeds? Second, *region-level stability*: do the same broader hotspot areas reappear across seeds even when the exact HH membership changes? All cross-seed comparisons in this section use the fixed-test-set protocol from Section 3.2, so that observation identities are aligned across runs.

4.3.1 Point-level stability: HH selection frequency

Let f_i denote the fraction of random seeds (out of 10) in which test point i is classified as HH under the global Rashomon set (top-$K = 25$) and the default neighborhood definition. Figure 1 visualizes this per-point HH frequency in PCA space.
At the point level, stability is limited. Only about 3.2% of the fixed test set satisfies $f_i \geq 0.8$, and only about 6.1% satisfies $f_i \geq 0.5$. More broadly, the total *HH support*—the set of observations classified as HH in at least one seed—covers about 21.6% of the fixed test set. Within this support, HH membership is typically infrequent: the mean frequency

is 0.36, the median is 0.20, and about 56.3% of support points lie in the lowest frequency bucket $(0, 0.2]$. Defining a recurrent *HH core* as points with $f_i \geq 0.5$, the core-to-support ratio is about 28.3%, indicating that only a minority of ever-detected HH points recur in at least half of the runs.

The main conclusion is therefore that **individual observations are not stably labeled as hotspots**. Exact HH membership varies substantially across seeds. At the same time, this variability is not spatially diffuse: higher-f_i values remain concentrated in one part of the PCA plane rather than being scattered uniformly. Thus, even though pointwise hotspot assignment is unstable, the broader location of multiplicity remains informative.

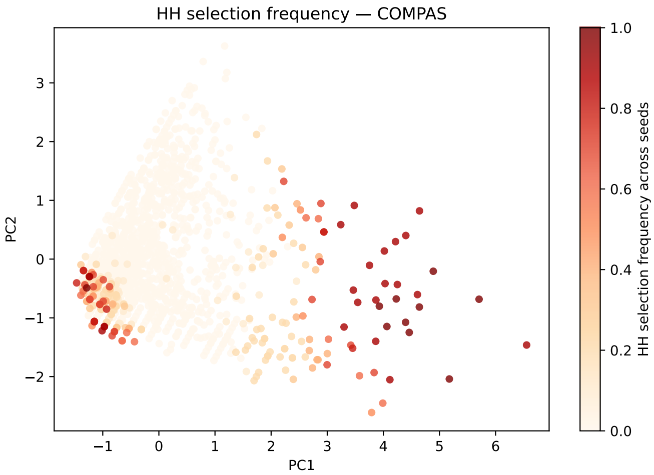


Figure 1: Per-point HH frequency f_i across 10 seeds on the fixed COMPAS test set. Color encodes how often each observation is classified as HH. Most observations have low HH frequency, but the higher-frequency points remain localized in feature space.

4.3.2 Overlap of exact HH sets

To quantify agreement on the exact set of hotspot observations, we compare the HH masks between pairs of seeds using the Jaccard index

$$J_{ss'} = \frac{|H_s \cap H_{s'}|}{|H_s \cup H_{s'}|}.$$

A value of 0 indicates no overlap between two HH sets, whereas a value of 1 indicates identical HH sets.

Figure 2 shows the resulting overlap matrix for variance-based HH masks on COMPAS. The mean off-diagonal Jaccard similarity is about 0.36, with values ranging from approximately 0.11 to 0.63 across seed pairs. This should be interpreted as moderate overlap. It is far below perfect agreement, confirming that exact HH membership changes substantially across seeds. At the same time, it is much larger than the overlap expected from unrelated sparse HH masks: for an average HH rate of about 7.8% in the fixed-test stability runs, an independent random-overlap baseline would be roughly 0.04. Thus, exact HH membership is not highly stable, but it is also not random.

This set-level analysis adds a second view to the pointwise frequency results from the previous subsection. The frequency analysis asks how often each individual observation is classified as HH, whereas the Jaccard analysis asks how similar the complete HH masks are between pairs of runs. Both views lead to the same conclusion: exact HH membership is only partly recurrent and remains sensitive to seed-specific variation. However, this should not be interpreted as evidence against spatial structure. Jaccard overlap is sensitive to boundary variation, so modest shifts at the edges of a hotspot can reduce overlap even when the broader hotspot area remains similar. This motivates the component-level regional analysis in the next subsection.

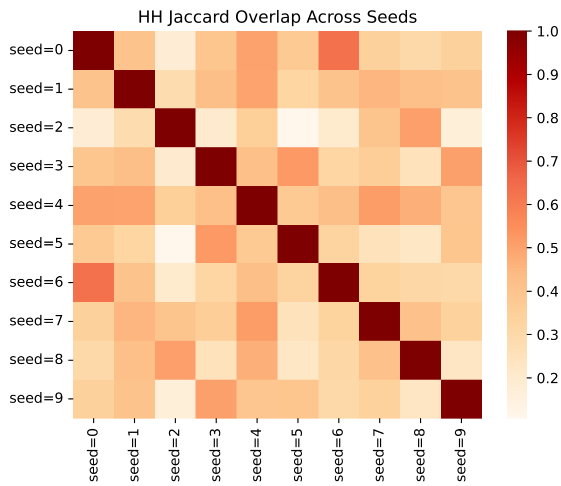


Figure 2: Pairwise Jaccard overlap of variance-based HH masks between seeds on the fixed COMPAS test set. Diagonal entries equal one by construction; off-diagonal values show moderate but variable agreement in exact HH membership across runs.

4.3.3 Regional structure: connected HH components

To move beyond individual HH points, we extract connected components of HH nodes in the same kNN graph used for LISA, discarding components smaller than 5 points. This analysis asks whether detected HH observations form coherent graph regions or merely appear as isolated points.

Table 6 summarizes the component statistics across runs. On COMPAS, HH points typically form about 3–4 connected components per run, with one dominant component whose maximum size averages about 62.5 test points (median 62). Hotspots therefore appear as a small number of contiguous regions rather than as isolated singletons.

Table 6: HH connected-component statistics across runs on COMPAS (minimum component size 5).

Dataset	Mean n components	Mean max comp. size	Median max comp. size	Max (over runs)
COMPAS	3.50	62.50	62	98

We then compare these regions across seeds. For each pair of runs, each component in one run is matched to the most similar component in the other run using Jaccard overlap on the corresponding fixed test observations. The mean best-match component Jaccard is about 0.35. This should be interpreted as moderate rather than strong recurrence. It is far below perfect agreement, indicating that component boundaries and memberships vary across seeds, but it is also not negligible for sparse hotspot regions.

The main contribution of the component analysis is therefore not to show that hotspot regions are fixed across runs. Rather, it shows that when HH points are detected, they often form contiguous graph regions. This supports a regional interpretation of multiplicity and motivates the component-level rule extraction in Section 4.7.

Taken together, the stability analysis shows that exact HH labels vary across seeds, while HH observations still tend to form contiguous graph regions. On COMPAS, Moran's I remains positive in all runs, but both HH counts and hotspot boundaries vary substantially. Predictive multiplicity should therefore be interpreted as a regional phenomenon with partially recurring structure, rather than as a fixed pointwise-stable labeling.

4.4 Robustness to Analytical Choices

We next assess whether the main spatial conclusions depend on several key analytical choices: the Rashomon set size K, the neighborhood size k used in the feature-space graph, post-hoc probability calibration, and the precise graph construction itself. Across all four checks, the absolute magnitude of disagreement and the extent of detected hotspot regions can change, but the central conclusion remains stable: predictive multiplicity continues to show positive spatial clustering under a wide range of reasonable specifications.

4.4.1 Sensitivity to Rashomon Set Size K

We first vary the Rashomon set size over $K \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ using global top-K selection. Figure 3 shows the resulting sensitivity curves for COMPAS; the corresponding plots for German Credit and Adult are reported in Appendix F.

As expected, the overall magnitude of disagreement increases with K. Mean predictive variance rises as additional near-optimal models are included. In contrast, the spatial organization of disagreement changes less strongly. On COMPAS, Moran's I varies most at small K and then remains in a comparatively narrow range once K reaches roughly 20–25; the number of HH hotspots also changes less than the mean variance. German Credit and Adult show broadly similar aggregate behavior: predictive variance grows as the Rashomon set widens, while the spatial pattern changes less than the raw scale of disagreement. For German Credit, hotspot-count comparisons remain less stable because the number of HH observations is very small.

The main implication is that larger Rashomon sets primarily add *more* disagreement, not a fundamentally different *structure* of disagreement. This supports the use of $K = 25$ as a pragmatic default: it is large enough to capture the main observed spatial organization of predictive multiplicity, while avoiding a much larger model set for which the qualitative pattern changes only modestly in the sensitivity analysis.

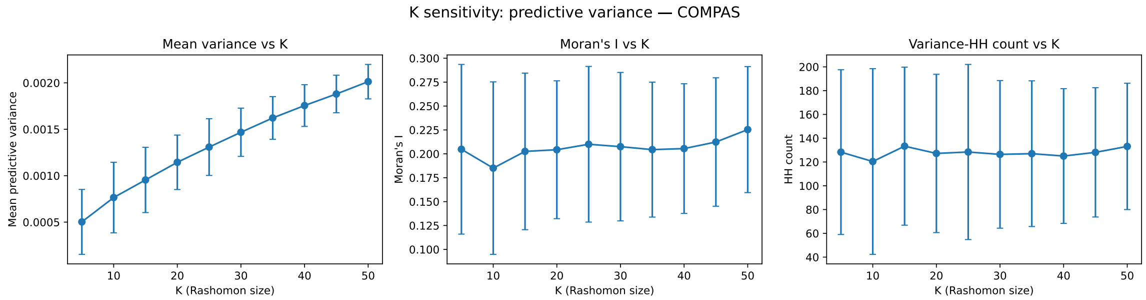


Figure 3: Sensitivity to Rashomon set size K on COMPAS (mean $\pm$ std over runs). Panels show mean predictive variance, Moran's I, and variance-based HH count.

4.4.2 Sensitivity to kNN Neighborhood Size k

We next vary the neighborhood size k used in the feature-space graph and recompute the variance-based spatial metrics. Figure 4 shows the resulting curves for COMPAS; the corresponding curves for German Credit and Adult are reported in Appendix F.

Mean predictive variance is invariant to k, since it is defined before the graph is constructed. By contrast, Moran's I decreases as k grows, reflecting the smoothing effect of larger neighborhoods. On COMPAS, Moran's I falls from roughly 0.29 at small k to about 0.12 at $k = 100$. At the same time, HH counts tend to increase with larger neighborhoods on COMPAS and Adult, because broader neighborhoods can merge local structure into larger statistically significant regions. German Credit shows the same direction of effect, though at a much smaller absolute scale.

Crucially, the clustering signal remains positive throughout the examined range. Thus, while the precise extent of detected hotspots depends on the neighborhood definition, the existence of spatial structure does not. The chosen defaults ($k = 30$ for COMPAS and German Credit, $k = 60$ for Adult) therefore represent a reasonable compromise between locality and graph stability rather than a finely tuned requirement for obtaining positive results.

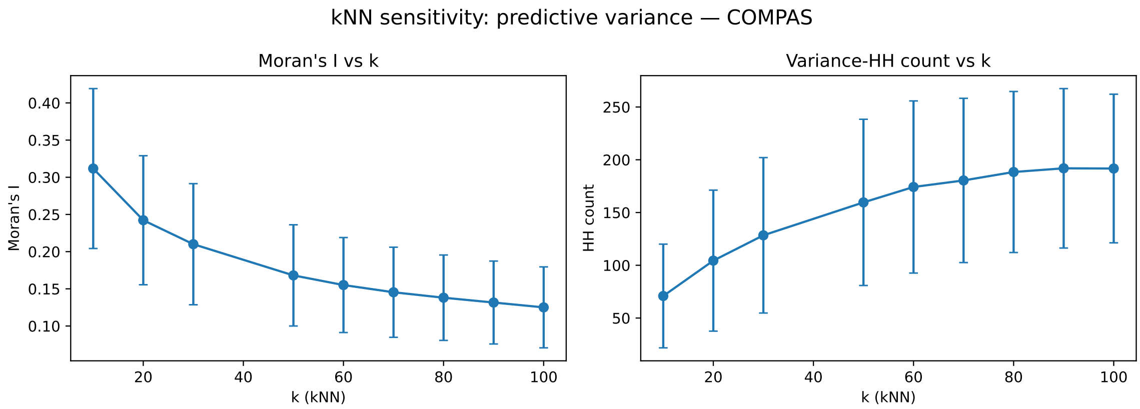


Figure 4: Sensitivity to kNN neighborhood size k on COMPAS (mean $\pm$ std over runs). Panels show Moran's I and variance-based HH count. Moran's I decreases with increasing k, while HH counts change with the neighborhood definition.

4.4.3 Calibration Robustness

To test whether the observed hotspot structure is primarily driven by the probability scale of the models, we apply post-hoc calibration to each model in the global Rashomon set and then recompute predictive variance and spatial metrics on the calibrated predictions. Figure 5 summarizes the resulting run-level changes.

Calibration affects the magnitude of disagreement and the exact HH assignments, but it does not eliminate the spatial structure of multiplicity. On COMPAS, Platt scaling reduces mean variance and slightly lowers Moran's I, while isotonic calibration increases mean variance but also lowers Moran's I on average. Thus, calibration changes the scale of the variance field, but the spatial clustering signal remains positive. On German Credit, the effects are harder to interpret because the number of HH observations is very small; small absolute changes in HH membership can therefore produce large relative changes in overlap statistics. On Adult, calibration tends to increase HH counts, while changes in Moran's I remain modest in absolute terms.

The Jaccard overlap between pre- and post-calibration HH masks shows that exact hotspot membership is only partially preserved. This is expected, because calibration can shift predicted probabilities and therefore alter local variance values near the LISA significance threshold. The overlap results should therefore be interpreted as evidence that hotspot boundaries are sensitive to calibration, not as evidence that the exact same HH points are recovered after calibration.

Overall, the calibration analysis supports a moderate robustness conclusion. Post-hoc calibration changes the scale of disagreement and the precise hotspot masks, but it does not remove the positive spatial autocorrelation of predictive multiplicity. This complements work that studies calibration as a way to reduce aggregate predictive multiplicity (Cavus, 2026): in the present analysis, calibration may change the magnitude and boundaries of hotspots, but the qualitative spatial signal remains visible. Thus, the main spatial findings do not appear to be solely an artifact of uncalibrated probability scores, although exact hotspot membership remains calibration-sensitive.

4.4.4 Alternative Graph Constructions

Finally, we test whether the spatial results depend strongly on the feature-space representation used to construct the graph. In addition to the baseline Euclidean kNN graph, we consider a graph built in PCA-reduced space (5 components) and a cosine-distance graph. Table 7 reports the resulting Moran's I and HH counts.

The resulting Moran's I values remain positive across the tested graph constructions, although the extent of detected HH regions changes with the graph definition. The PCA-reduced graph provides a compressed representation of the preprocessed feature space, while the cosine-distance graph changes the notion of local similarity directly. Differences across these alternatives therefore indicate that exact HH counts and hotspot boundaries are graph-dependent. However, the persistence of positive Moran's I across the tested constructions supports the conclusion that predictive multiplicity is not merely an artifact of the baseline Euclidean graph definition.

The key cross-dataset pattern remains similar across graph constructions: COMPAS shows the strongest localized spatial structure, German Credit shows a weaker signal

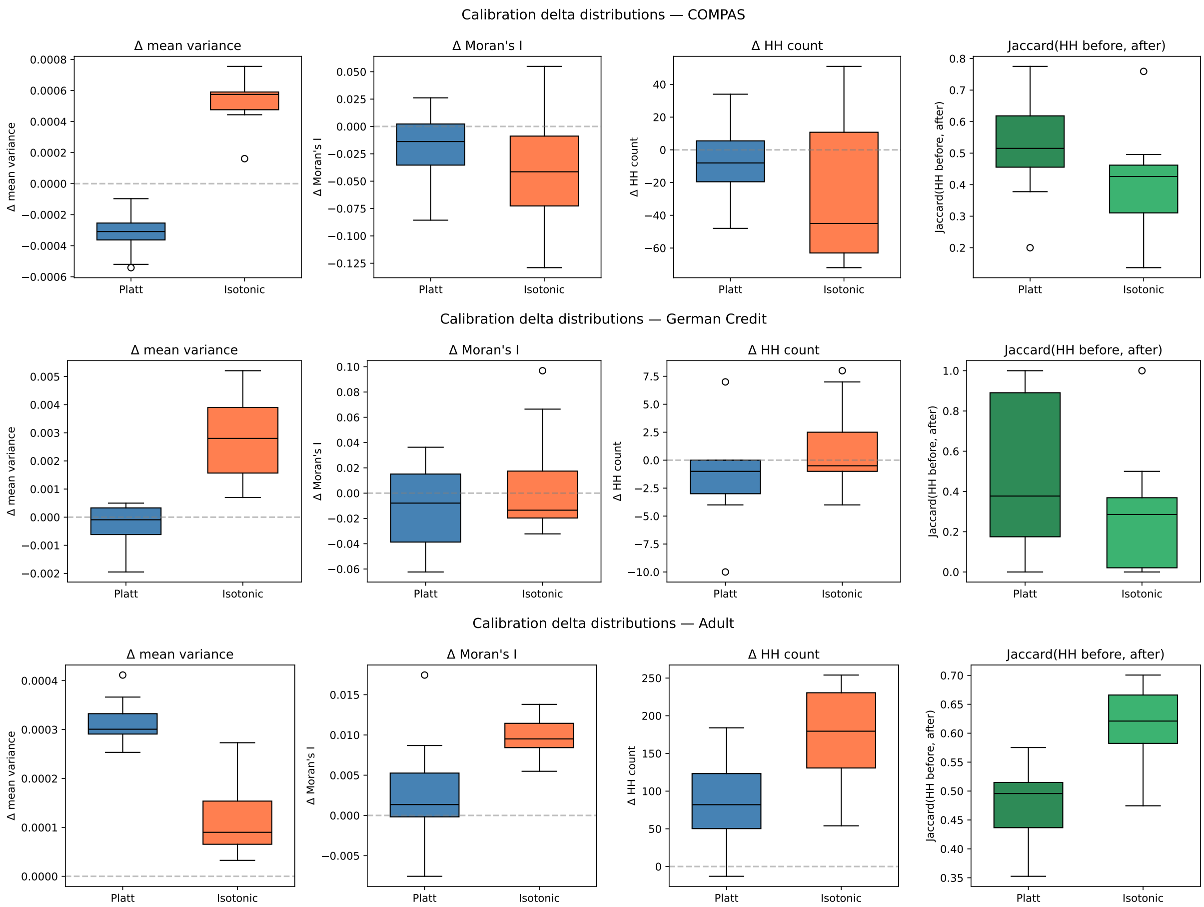


Figure 5: Calibration robustness across runs. From top to bottom: COMPAS, German Credit, and Adult. For each dataset, the panels show the change after calibration in mean variance, Moran's I, and HH count, together with the Jaccard overlap between HH masks before and after calibration.

Table 7: Spatial metrics under alternative kNN graph constructions (mean $\pm$ std over 10 seeds; dataset-specific default k: 30 for COMPAS and German Credit, 60 for Adult).

Dataset	Method	Moran's I	HH count
COMPAS	Euclidean (baseline)	0.210 ± 0.081	128.4 ± 73.6
	PCA (5 comp.)	0.212 ± 0.085	140.4 ± 73.9
	Cosine	0.213 ± 0.087	149.3 ± 73.2
German Credit	Euclidean (baseline)	0.088 ± 0.039	5.4 ± 5.4
	PCA (5 comp.)	0.051 ± 0.035	4.7 ± 5.5
	Cosine	0.096 ± 0.041	7.9 ± 6.4
Adult	Euclidean (baseline)	0.075 ± 0.022	631.0 ± 239.7
	PCA (5 comp.)	0.073 ± 0.023	694.3 ± 239.9
	Cosine	0.080 ± 0.022	698.4 ± 217.7

with few HH observations, and Adult shows broader but more diffuse hotspot regions.
Taken together, the robustness checks show a consistent pattern. Analytical choices can affect the *magnitude* of disagreement, the size of detected hotspot regions, and the exact HH assignments, but they do not remove the positive spatial clustering signal. The qualitative finding that predictive multiplicity shows positive spatial clustering therefore persists across the tested variations in Rashomon set size, neighborhood size, calibration, and graph construction. The magnitude of this signal and the resulting HH regions remain dataset-dependent, with German Credit showing the weakest and least stable hotspot structure.

4.5 Model-Family and Hyperparameter Associations

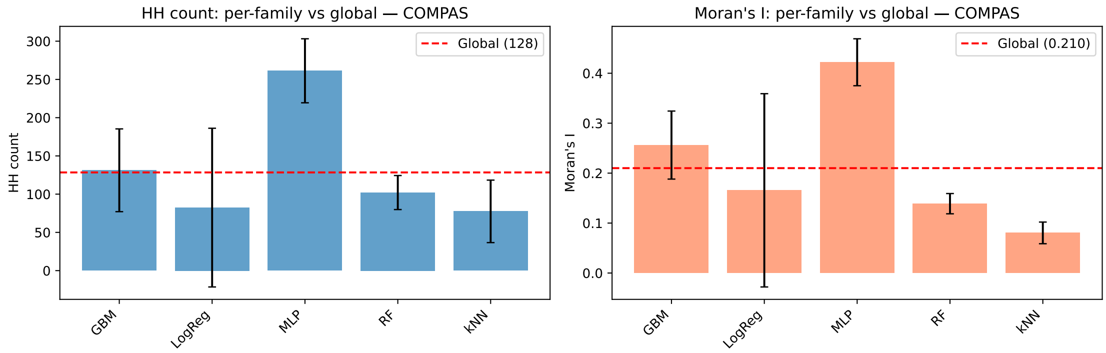
We now analyze which modeling choices are associated with predictive multiplicity within the sampled candidate pool. The analysis proceeds in four steps. First, we compare global and per-family Rashomon sets to assess whether pooling across model families hides family-specific spatial structure. Second, we decompose predictive variance into between-family and within-family components. Third, we identify which grouped hyperparameter values account for variation in the model-level disagreement contribution V_m within each family. Finally, we examine whether these descriptive associations change inside HH hotspot regions.
All attribution results in this section are descriptive rather than causal. They summarize how disagreement varies across the trained and retained candidate models, but they do not estimate the causal effect of changing one hyperparameter while holding all other factors fixed. An auxiliary performance-control diagnostic is reported in Appendix C.

4.5.1 Global vs. Per-Family Rashomon Sets

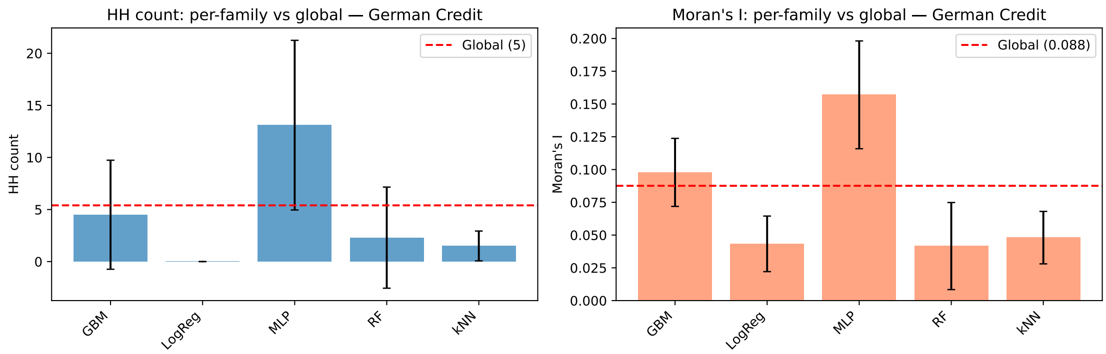
We compare two ways of constructing the Rashomon set: a *global* set, defined as the top-$K = 25$ models selected across all model families, and *per-family* sets, defined as the top-$K = 25$ models selected separately within each family. The global construction reflects disagreement among all near-optimal models taken together, whereas the per-family construction isolates the spatial structure that appears within a given model class. The key question is whether pooling across families obscures family-specific multiplicity patterns.
Table 8 gives the per-family breakdown for COMPAS together with the corresponding global reference row, denoted *Global (all families)*. Figure 6 compares per-family HH counts and Moran's I to the global reference values across all three datasets.

COMPAS. COMPAS shows the clearest family-level heterogeneity. According to Table 8, MLP has by far the strongest spatial structure, with the largest Moran's I and the largest HH count, both well above the global reference. kNN exhibits the highest mean variance but the weakest Moran's I among the families, indicating that large disagreement does not necessarily translate into strongly localized clustering. Logistic regression lies at the opposite extreme, with very small mean variance and comparatively weak hotspot structure, albeit with high variability across seeds. GBM and random forest occupy an intermediate position. Figure 6(a) makes the main point visually clear: several per-family

(a) COMPAS



(b) German Credit



(c) Adult

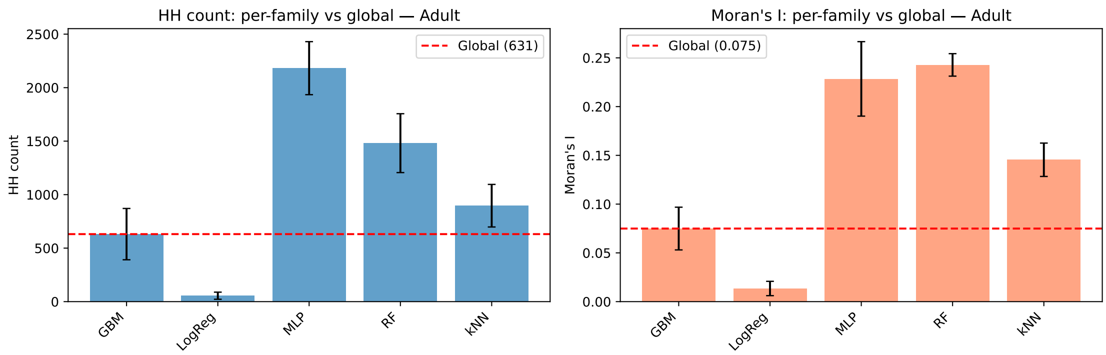


Figure 6: Family importance within the Rashomon set. Bars show the mean share of total predictive variance attributable to between-family differences for all observations, HH observations, and non-HH observations. The remaining share corresponds to within-family variation. Error bars show standard deviations over seeds.

Table 8: Per-family Rashomon sets and global reference on COMPAS. Per-family rows use the top-$K = 25$ models within each family. The *Global (all families)* row uses the top-$K = 25$ models selected from the full candidate pool across all families. Values are mean ± std over runs. HH rate reports the share of held-out test observations classified as HH.

Selection	Mean variance (mean ± std)	Moran's I (mean ± std)	HH count (mean ± std)	HH rate (mean ± std)
Global (all families)	0.001308 ± 0.000306	0.210 ± 0.081	128.4 ± 73.6	8.9% ± 5.1%
GBM	0.001245 ± 0.000290	0.256 ± 0.068	131.2 ± 54.1	9.1% ± 3.7%
LogReg	0.000060 ± 0.000128	0.166 ± 0.193	82.4 ± 103.8	5.7% ± 7.2%
MLP	0.001256 ± 0.000133	0.422 ± 0.047	261.3 ± 41.9	18.1% ± 2.9%
RF	0.003239 ± 0.000355	0.139 ± 0.020	102.1 ± 22.2	7.1% ± 1.5%
kNN	0.010692 ± 0.001397	0.080 ± 0.022	77.5 ± 40.9	5.4% ± 2.8%

signals exceed the global reference, especially for MLP, so the pooled global Rashomon set understates the strength of family-specific hotspot structure.

German Credit and Adult. The same comparison is informative on the other datasets, but with different emphases. On German Credit, the absolute HH counts remain small across all constructions, so family differences should be interpreted cautiously in absolute terms. Nonetheless, Figure 6(b) still shows that the spatial signal is not uniform across families. On Adult, the heterogeneity is much more pronounced: MLP and random forest stand out with markedly larger Moran's I and HH counts, whereas logistic regression remains comparatively low on both measures. Thus, as on COMPAS, the pooled global construction compresses family-level structure that is clearly visible when families are examined separately.

Taken together, these comparisons support two conclusions. First, the model families associated with stronger structured disagreement are dataset-dependent rather than fixed across problems. Second, global pooling can mask substantial family-specific multiplicity patterns, particularly on COMPAS and Adult. This motivates the variance-decomposition analysis in the next subsection, where disagreement is summarized more directly in terms of between-family and within-family variation.

4.5.2 Family Importance

Model family accounts for a substantial share of observed predictive multiplicity across all datasets. Figure 7 shows the between-family and within-family components of predictive variance within the Rashomon set, separately for all observations, HH hotspots, and non-HH observations. The between-family component is large in all three datasets, indicating that disagreement among near-optimal models is strongly associated with broad model-class differences rather than only with variation within a fixed family.

The strongest family association appears on Adult, where the between-family share is about 0.71 over all test observations and increases to about 0.79 inside HH regions. COMPAS shows a lower but still substantial family contribution, increasing from about 0.41

over all observations to about 0.52 inside HH regions. German Credit has a similarly high all-observation family share of about 0.53, but the HH and non-HH estimates are very similar; because German Credit contains only few HH observations, its hotspot-specific decomposition should be interpreted cautiously.

Comparing the Rashomon set with the full candidate pool on all observations shows that family structure is not simply a property of the entire search space. On Adult, family importance is much larger in the Rashomon set than in the full pool (0.71 vs. 0.36), and German Credit shows the same pattern (0.53 vs. 0.30). On COMPAS, by contrast, the Rashomon and full-pool values are very similar (0.41 vs. 0.42). Thus, the role of model family depends on the dataset: in Adult and German Credit, near-optimal selection sharpens between-family differences, whereas on COMPAS family-level disagreement is already visible in the broader candidate pool.

Taken together, these results support the interpretation that predictive multiplicity is partly structured by model-family choice. This association is especially pronounced in Adult and remains substantial on COMPAS and German Credit. Moreover, for Adult and COMPAS, the family contribution increases inside HH regions, suggesting that hotspot predictions are particularly sensitive to the choice among model classes.

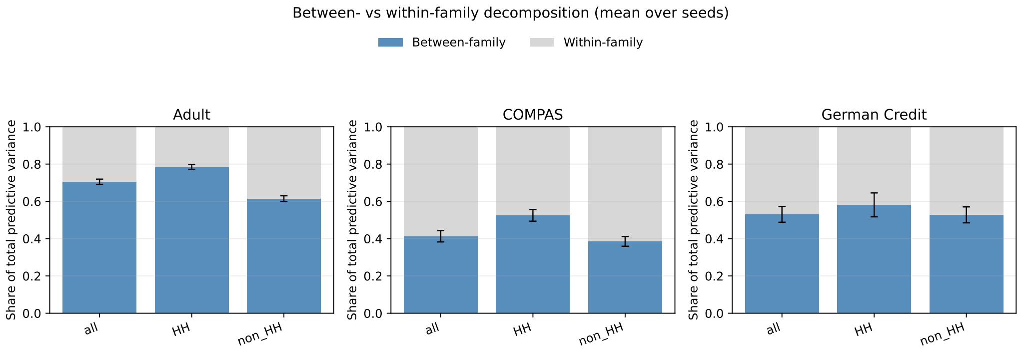


Figure 7: Family importance within the Rashomon set. Bars show the mean share of total predictive variance attributable to between-family differences, with the remaining share corresponding to within-family variation. Error bars show standard deviations over seeds.

4.5.3 Within-Family Hyperparameter Associations

After separating between-family from within-family variation, we next ask which grouped hyperparameter values account for variation in the remaining disagreement inside each model family. For each family-specific Rashomon set, we compute the model-level disagreement contribution V_m and decompose variation in V_m by hyperparameter groups. Categorical and low-cardinality numeric hyperparameters are grouped by exact value, while high-cardinality numeric hyperparameters are grouped into tertiles. Hyperparameters with insufficient group support are excluded for the corresponding seed.

Figure 8 shows the resulting within-family hyperparameter importance values. The main pattern is that within-family disagreement is not diffuse across all hyperparameters. Instead, variation in V_m is often concentrated in a small number of grouped hyperparameter values. For logistic regression, variation across grouped values of regularization strength C accounts for most of the observed variation in V_m within retained logistic-regression models, while the elastic-net mixing parameter l_1 ratio contributes much less. For kNN, variation across the number of neighbors is the dominant association in every dataset, indicating that the locality scale of the classifier is closely related to its multiplicity behavior.

Tree-based and neural-network families show more distributed but still interpretable patterns. For GBM, learning rate and tree-depth related parameters are consistently among the largest associated sources of within-family variation, although their ordering differs across datasets. Random forests are more dataset-dependent: maximum depth is largest on COMPAS, while depth, feature subsampling, and leaf-size related parameters all contribute on German Credit and Adult. For MLPs, learning-rate initialization and hidden-layer size show the strongest associations, with regularization and activation contributing more moderately.

Overall, these results show that within-family multiplicity is structured by identifiable modeling choices rather than by random variation across the hyperparameter search space. Simpler families are often dominated by one complexity-related parameter, whereas more flexible families distribute variation across several architectural, regularization, or optimization-related parameters. These patterns should be interpreted as descriptive associations within the sampled Rashomon approximations, not as causal hyperparameter effects.

4.5.4 Hotspot-Specific Hyperparameter Associations

The preceding analysis identifies which grouped hyperparameter values are associated with within-family disagreement on average. We next examine whether these associations change inside multiplicity hotspots. For each hyperparameter, we compare its V_m-based importance on HH observations with its importance on all test observations:

$$\Delta\rho(h) = \rho_{\mathrm{HH}}(h) - \rho_{\mathrm{all}}(h).$$

Positive values indicate that variation across grouped values of hyperparameter h accounts for a larger share of within-family disagreement inside HH regions, whereas negative values indicate a lower association inside hotspots.

Table 9 reports the largest positive and negative shifts for COMPAS, which is the clearest real-data case of localized hotspot structure. The results suggest that hyperparameter associations are not spatially uniform. For GBM, subsampling has a larger association with V_m inside HH regions, while the number of estimators has a slightly smaller association. For MLP, learning-rate initialization shows the largest positive shift, suggesting that optimization-related settings are more strongly associated with disagreement inside the most unstable regions. In contrast, random forest depth and kNN weighting show weaker associations inside HH regions. Logistic regression shows no positive shift, indicating that its already dominant regularization association does not become stronger inside COMPAS hotspots.

These shifts should be interpreted descriptively rather than causally. They indicate that the relative sources of within-family disagreement can differ between hotspot and non-hotspot regions, but several estimates have substantial cross-seed variability. HH-specific

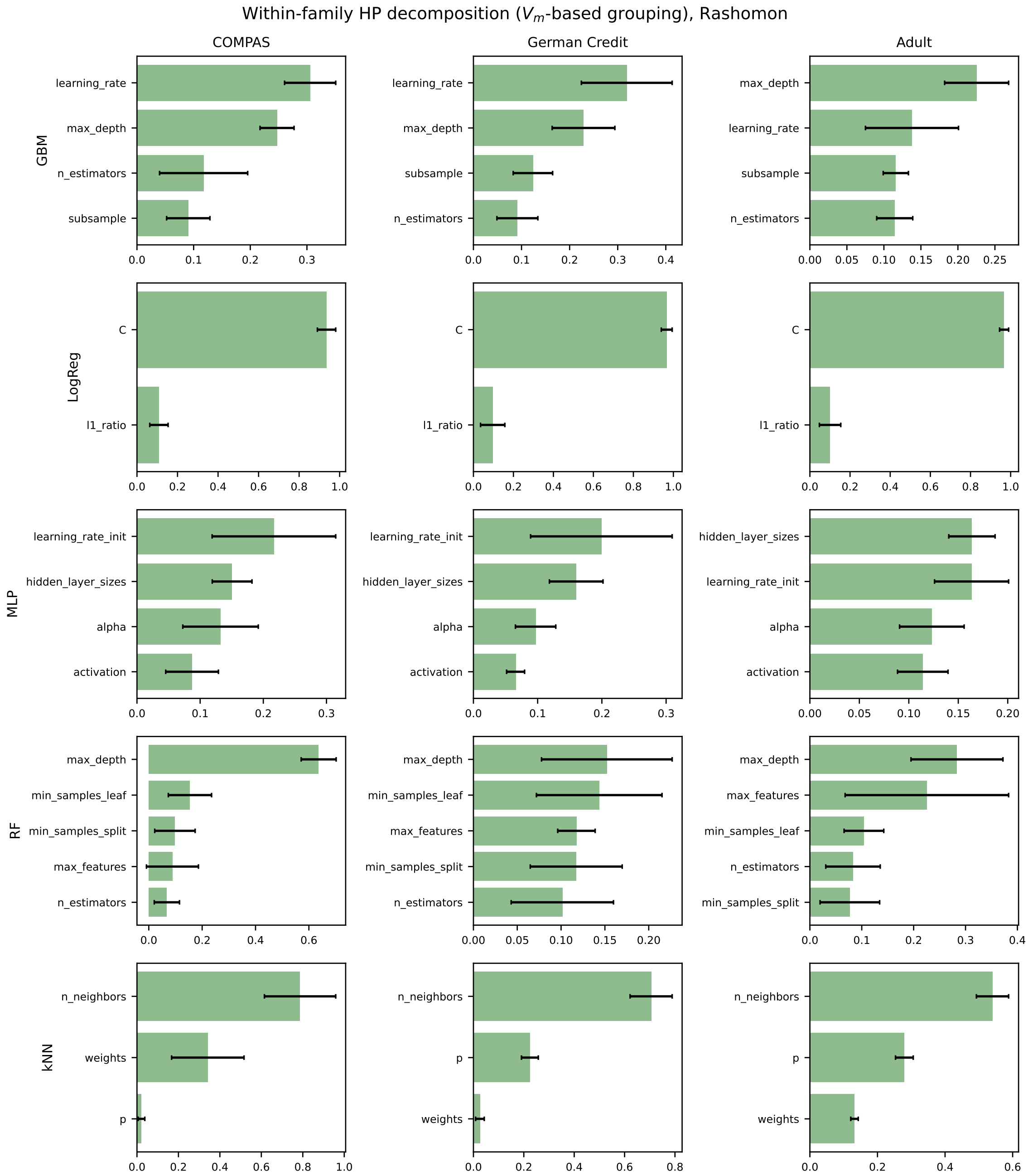


Figure 8: Within-family hyperparameter importance within the Rashomon set. Bars show mean V_m-based importance over seeds, and error bars show standard deviations. Hyperparameters are grouped by exact value for categorical and low-cardinality numeric parameters and by tertiles for high-cardinality numeric parameters.

estimates for datasets with very small hotspot regions, especially German Credit, are therefore treated as auxiliary diagnostics rather than primary evidence.

Table 9: Largest positive and negative changes in within-family hyperparameter importance inside COMPAS HH hotspots relative to all observations. Positive values indicate greater V_m-based importance inside HH regions. The kNN negative shift is based on six valid seeds because of group-support filtering.

Family	Largest positive HH shift	Largest negative HH shift
GBM	subsample (+0.021)	max_depth (−0.014)
LogReg	none	l1_ratio (−0.004)
MLP	learning_rate_init (+0.042)	hidden_layer_sizes (−0.002)
RF	min_samples_split (+0.007)	max_depth (−0.045)
kNN	p (+0.006)	weights (−0.048)

As an auxiliary performance-control diagnostic, we also test whether the observed hyperparameter associations mainly proxy for validation Brier score. The analysis suggests that validation performance accounts for more model-level disagreement in the full candidate pool than inside the Rashomon set, while non-performance hyperparameters remain important within the Rashomon set. Details are reported in Appendix C.

4.6 Synthetic Data Validation

To validate the proposed pipeline, we evaluate it on synthetic datasets with known ground-truth ambiguous regions. These experiments test whether LISA-based hotspot detection can recover regions of inherent predictive uncertainty.

4.6.1 Single-Island Design

The single-island synthetic dataset consists of $n = 3000$ observations in a two-dimensional feature space (x_1, x_2). It contains two well-separated Gaussian clusters that form stable regions ($n = 2400$ in total) and one ambiguous island ($n = 600$), generated as a uniform disk with $P(y = 1 \mid x) \approx 0.5$. We apply the standard experimental pipeline with five model families (LogReg, kNN, RF, GBM, MLP), 50 candidates per family (250 total), and a global top-$K = 25$ Rashomon set. Evaluation is performed on a held-out test set of size 600.

Figure 9 shows the data-generating process and the ground-truth ambiguous region. Figure 10 shows the resulting pointwise predictive variance together with the detected LISA HH hotspots. The variance map reveals that multiplicity is concentrated in the central island rather than in the two stable Gaussian regions, and the HH mask closely follows this structure.

This spatial concentration is also reflected in the global autocorrelation statistic. The observed Moran's I is 0.2536, indicating clear positive spatial clustering of predictive variance. Under the prediction-permutation null, the null mean is -0.0024 with standard deviation 0.0098, and the one-sided empirical p-value is 0.0099. Thus, the observed

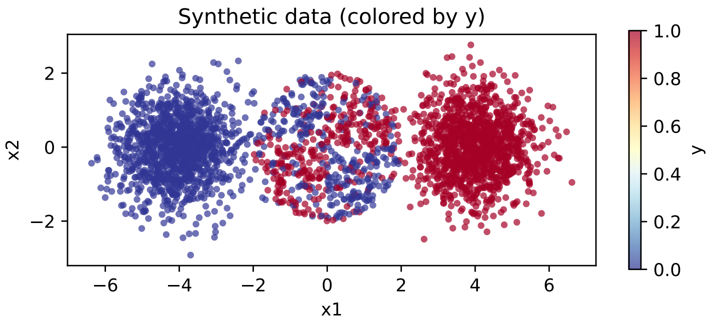


Figure 9: Synthetic single-island dataset. Two stable Gaussian clusters (left/right) and one ambiguous island (center, dashed circle) with $P(y = 1 \mid x) \approx 0.5$.

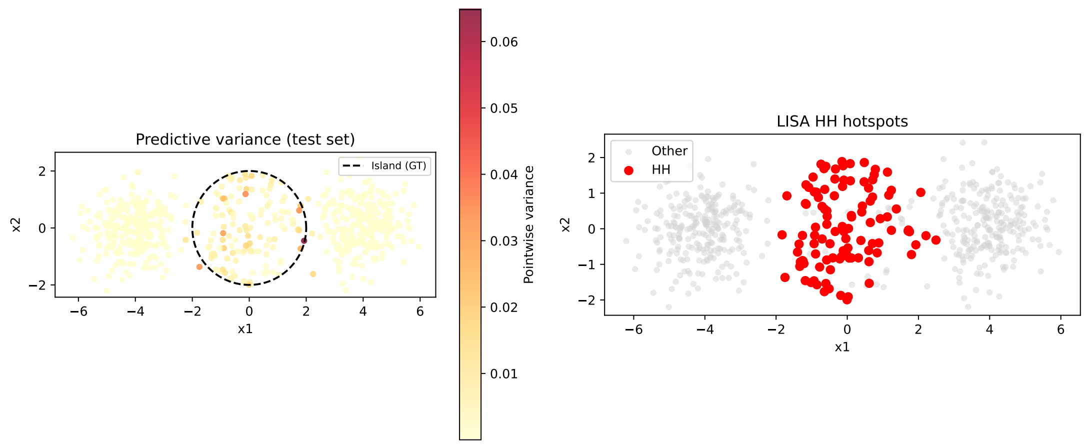


Figure 10: Left: Predictive variance across the Rashomon set. Right: LISA HH hotspots (red). The detected hotspot region aligns with the ground-truth ambiguous island.

clustering is far larger than expected when predictions are detached from feature-space locations.

At the local level, LISA identifies 99 HH points and 405 LL points. Relative to the ground-truth island, which contains 120 test observations, this yields a precision of 0.9697, a recall of 0.8000, and a Jaccard overlap of 0.7805. In absolute terms, 96 of the 99 detected HH points lie inside the ambiguous island. This shows that the detected hotspot is highly specific to the intended ambiguous region, while still recovering most of its extent. Sensitivity to the LISA false-discovery-rate threshold is reported in Appendix E; across the tested thresholds, the detected HH region remains stable and continues to align with the ground-truth island.

Overall, the single-island experiment provides a controlled validation of the pipeline. It shows that when predictive ambiguity is concentrated in one coherent region, the combination of pointwise variance, Moran's I, and LISA recovers that region accurately and with strong statistical support.

4.6.2 Three-Islands Design

The second synthetic design consists of $n = 5000$ observations in a two-dimensional feature space (x_1, x_2). It contains two stable Gaussian clusters ($n = 3400$), three ambiguous islands ($n = 1500$ in total), and an additional set of 100 outliers. The outliers are constructed in two groups: low-multiplicity outliers placed in regions with stable class probability, and high-multiplicity outliers placed in small ambiguous regions. This makes it explicit which outliers should and should not be detected as hotspots. Figure 11 shows the data-generating process, while Figure 12 shows the resulting predictive variance and detected LISA HH hotspots.

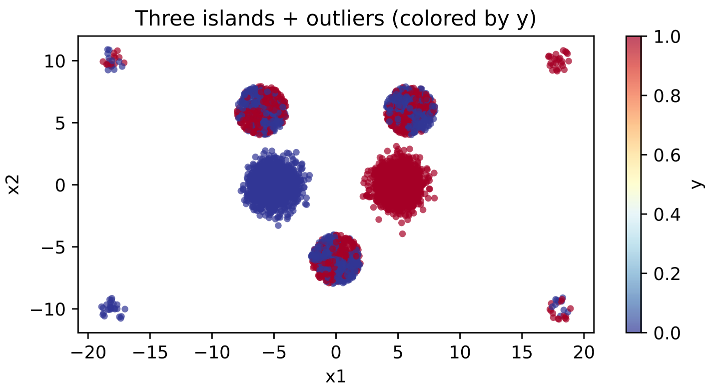


Figure 11: Synthetic three-islands dataset. Two stable Gaussian clusters, three ambiguous islands, and two types of outliers: stable low-multiplicity outliers and ambiguous high-multiplicity outliers.

Applying the standard pipeline yields 268 HH points. The variance map shows that predictive multiplicity is concentrated in the three ambiguous islands and in the designed high-multiplicity outliers, while the stable Gaussian clusters and low-multiplicity outliers remain largely outside the hotspot region. This makes hotspot detections outside the

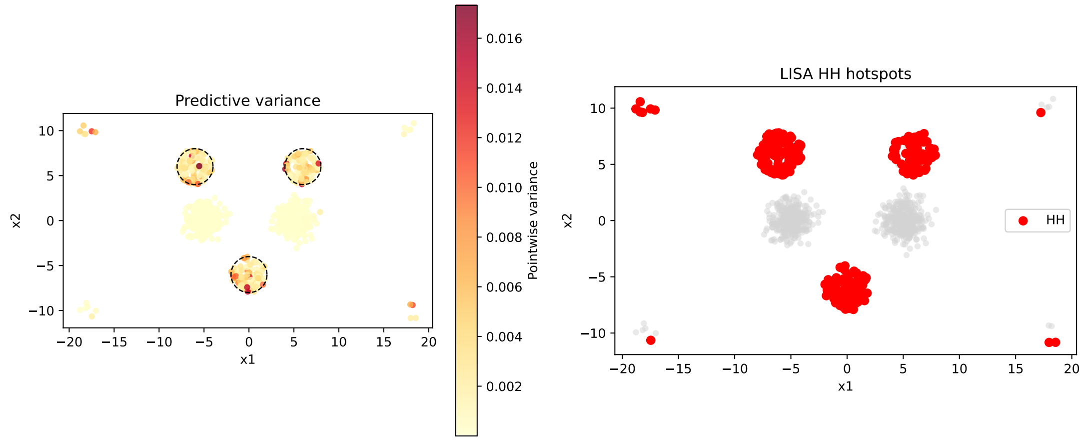


Figure 12: Left: Predictive variance across the Rashomon set. Right: LISA HH hotspots (red). The detected hotspots align with the three ambiguous islands and additionally recover the designed high-multiplicity outliers, while the stable outliers are not systematically selected.

main islands directly interpretable, since they can be compared against the known outlier construction.

At the local level, recovery of the three island regions is strong. Relative to the ground-truth islands, which contain 294 test observations in total, the HH mask achieves a precision of 0.9627, a recall of 0.8776, and a Jaccard overlap of 0.8487. In absolute terms, 258 of the 268 HH points lie inside one of the three ground-truth islands. The remaining HH points correspond to the intentionally designed high-multiplicity outliers rather than arbitrary false positives. Sensitivity to the LISA false-discovery-rate threshold is reported in Appendix E; across the tested thresholds, the detected HH structure remains stable.

This experiment shows that the pipeline can recover several disconnected regions of predictive ambiguity while also distinguishing between outliers that should and should not exhibit high multiplicity. The resulting hotspot structure is therefore not only accurate with respect to the three island regions, but also consistent with the known behavior of the outlier groups.

4.7 Interpretable Rules for High-Multiplicity Regions

To make multiplicity hotspots more actionable, we extract simple rules that describe HH regions (LISA High–High hotspots) in terms of the original input features. The goal is not to fully explain every hotspot point, but to assess whether localized regions of predictive multiplicity correspond to recognizable subpopulations.

4.7.1 Rule Extraction Approach

We use shallow decision trees (maximum depth 3) to extract human-readable rules. Trees are fitted on the original feature space, and positive leaf nodes are converted into conjunctive rules. Each rule is evaluated by support, purity, recall, and lift.

We apply this procedure in two ways. First, we fit global HH rules, where the target is membership in the full HH set of a given run. In this case, purity is defined as

$$P(\text{HH} \mid \text{rule region}),$$

and lift is the ratio between this purity and the HH base rate.
Second, we fit component-level rules, where each connected HH component is treated as the positive class and all other test observations as the negative class. In this case, purity is defined as

$$P(\text{component} \mid \text{rule region}),$$

and lift is measured relative to the base rate of the corresponding component. The component-level analysis is especially relevant because earlier results show that hotspots are better interpreted as localized regions than as pointwise-stable labels.

4.7.2 Results on COMPAS

COMPAS provides the clearest interpretable structure. The global HH rules reveal a strong support–purity trade-off: highly pure rules exist, but they typically describe small subgroups, whereas broader rules have lower purity. Figure 13 shows this relationship. This indicates that the full HH set is heterogeneous and cannot be summarized well by a single global rule.

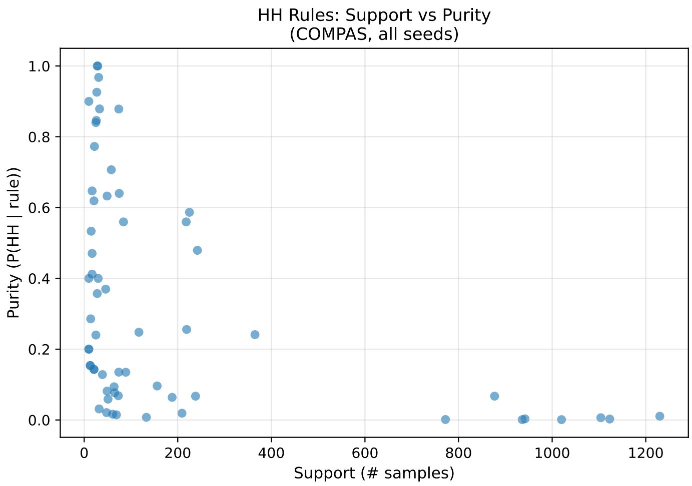


Figure 13: Support–purity trade-off for global HH rules on COMPAS, aggregated across seeds. High-purity rules exist, but they usually cover only small subgroups.

The component-level analysis gives a stronger regional interpretation. Across retained COMPAS hotspot components, shallow-tree rules often identify compact subregions with high enrichment. In the current analysis, 17 retained HH components across 8 seeds yield 36 component-level rules. Considering the highest-purity rule for each component, the median component size is 31 observations, the median purity is 0.53, the median recall is 0.96, and the median lift is 26.7. Thus, individual hotspot components are often more cleanly describable than the full global HH set.

Table 10 reports representative component-level rules. The strongest rules mostly involve age and prior offence count. For example, rules based on high values of `priors_count`, sometimes combined with higher age thresholds, identify compact regions with very high lift relative to the component base rate. A purity of 1.0 means that all observations satisfying the rule belong to the corresponding component, while recall below 1.0 indicates that the rule captures only a subset of the component, typically a clean core rather than the full region.
For readability, redundant conjuncts are simplified in the displayed rules; the reported support, purity, recall, and lift values are computed from the original tree leaves.

Table 10: Representative component-level HH rules on COMPAS. Rules are fitted separately for retained connected HH components.

Seed	Component	Rule	Support	Purity	Recall	Lift
6	1	priors_count > 17.5 AND age > 33	31	1.000	0.689	32.07
2	0	priors_count > 20.5	28	1.000	0.264	13.61
5	2	priors_count > 19.5 AND age > 36.5	27	0.926	0.833	44.54
7	1	priors_count > 16.5	39	0.923	0.857	31.71
8	0	priors_count > 14.5 AND age > 40.5	33	0.879	0.967	42.27

The feature-frequency results in Table 11 confirm this pattern. The dominant recurring descriptors are `priors_count` and `age`, which appear across most component rules and across several seeds. Race, sex, and charge-degree indicators occur only in a smaller number of component rules and should therefore be interpreted as component-specific rather than as global descriptors of hotspot membership.

Table 11: Feature frequency across component-level HH rules on COMPAS.

Feature	Rules	Seeds	Mean purity	Mean lift
age	55	10	0.343	10.10
priors_count	41	10	0.399	10.98
race_Caucasian	7	4	0.377	5.66
sex_Female	6	4	0.213	5.50
race_Other	6	2	0.227	10.11
c_charge_degree_F	5	2	0.377	17.73
c_charge_degree_M	3	2	0.280	10.83
sex_Male	2	2	0.357	14.84

We therefore interpret stability at the feature level rather than at the exact-rule level. The precise tree thresholds are not expected to be invariant across seeds or components, but the repeated use of the same variables indicates a robust descriptive pattern.
Overall, the rule analysis shows that HH regions are only partially interpretable at the global level, but substantially more interpretable once decomposed into connected hotspot components. This supports the main interpretation of the thesis: predictive multiplicity is localized and heterogeneous, but not arbitrary. It concentrates in recognizable regions of the feature space, especially regions characterized by age and prior criminal history.

4.8 Summary of Results

The experimental results show that predictive multiplicity is not merely a global disagreement summary, but a structured phenomenon with spatial, model-dependent, and interpretable components. Across datasets, model families, and robustness checks, several main patterns emerge.

1. **Spatial structure:** Predictive multiplicity is spatially clustered in feature space under the tested benchmark settings. Variance-based Moran's I is positive across all runs and statistically significant under the prediction-permutation null in all runs for COMPAS and Adult, and in 9 out of 10 runs for German Credit. Under this null, each model's marginal prediction distribution is preserved while predictions are randomly reassigned to test observations. The observed clustering therefore exceeds what would be expected if predictive variance were detached from feature-space location.

2. **Magnitude and concentration differ across datasets:** Disagreement magnitude and spatial concentration are not the same. German Credit exhibits the largest aggregate multiplicity, with the highest mean predictive variance (0.0050). Adult has weaker spatial concentration, but a larger absolute number of HH observations, reflecting a broader and more diffuse hotspot structure.

3. **Region-level stability despite point-level instability:** Exact HH membership varies across runs, but the fixed-test analysis on COMPAS shows moderate rather than negligible pointwise overlap, with mean pairwise Jaccard similarity around 0.36. Connected-component analyses further show that HH observations often form contiguous regions in the feature-space graph. Predictive multiplicity should therefore be interpreted as a regional phenomenon with partially recurring structure, while avoiding the stronger claim that individual observations have fixed hotspot labels.

4. **Model family is strongly associated with predictive variance:** Model family accounts for a substantial share of observed predictive variance, and this share increases inside hotspot regions on COMPAS and Adult. In the Rashomon set, family importance rises from about 0.41 to 0.52 on COMPAS and from about 0.71 to 0.79 on Adult when moving from all observations to HH regions. German Credit also shows high family importance, but its HH-specific estimate is less stable because very few HH points are detected.

5. **Within-family hyperparameters add structured variation:** Within model families, disagreement is not diffuse across all hyperparameters. Variation in V_m is often concentrated in a small number of grouped hyperparameter values, such as regularization strength C for logistic regression, the number of neighbors for kNN, learning-rate and tree-depth parameters for GBM, and learning-rate initialization or architecture choices for MLPs. These patterns should be interpreted as descriptive associations within the sampled Rashomon approximations rather than as causal hyperparameter effects.

6. **Robustness to analytical choices:** The main spatial conclusions are robust to reasonable changes in the analysis design. Increasing Rashomon set size K mainly increases the magnitude of disagreement, while the spatial pattern changes less strongly after moderate values of K. Increasing the neighborhood size k changes the smoothness and extent of detected regions, but does not remove positive spatial autocorrelation. Post-hoc calibration and alternative graph constructions likewise alter exact HH masks without eliminating the underlying clustering signal.

7. **Validation on synthetic data:** On synthetic datasets with known ambiguous regions, the pipeline recovers the intended instability regions with high precision and substantial recall. In the single-island design, LISA HH detection achieves precision 0.97, recall 0.80, and Jaccard overlap 0.78. In the three-islands design, it achieves precision 0.96, recall 0.88, and Jaccard overlap 0.85. This supports the validity of the spatial pipeline under controlled ground-truth conditions.

8. **Interpretability of hotspot regions:** High-multiplicity regions can be partly described by compact rules in the original feature space. Global HH rules show a strong support–purity trade-off, whereas component-level rules provide cleaner regional descriptions. On COMPAS, retained component-level rules repeatedly involve `priors_count` and age, with a median best-rule lift of 26.7 across retained components.

Overall, the findings support the central claim of this thesis: predictive multiplicity is a localized and structured property of the interaction between data geometry, model family, and hyperparameter choice. It concentrates in identifiable regions of feature space, remains visible under several robustness checks, and can be partially characterized by interpretable rules. Spatial multiplicity analysis therefore provides a useful diagnostic for identifying predictions that are especially sensitive to the choice among near-optimal models.

5 Discussion

This chapter interprets the empirical results beyond metric reporting, connects them back to the research questions, and discusses the practical and methodological implications of treating predictive multiplicity as a spatially structured phenomenon.

5.1 From Global Multiplicity to Local Risk

The central empirical message of this thesis is that predictive multiplicity is *localized*: it is not spread uniformly across the population, but concentrates in identifiable regions of feature space. Global disagreement metrics such as ambiguity, discrepancy, or disagreement rate remain useful because they quantify *how much* disagreement exists overall. However, by construction, they do not reveal *where* disagreement occurs or *which* individuals are most exposed to it. By combining observation-wise predictive variance with Moran's I and LISA, this thesis shows that disagreement is often organized into contiguous HH (High–High) regions—multiplicity hotspots—rather than appearing as a purely diffuse observation-level quantity.

This changes the practical interpretation of predictive reliability. Under a single deployed model, two individuals may receive similarly confident predicted probabilities, yet their predictions need not be equally reliable. One individual may lie in a stable region where near-optimal models agree closely, while the other may lie in a hotspot where equally accurate models disagree substantially. In that case, the second prediction is operationally less reliable even when its reported score appears equally confident. The key point is therefore not only *how uncertain* a prediction is, but also *how sensitive it is to the choice among near-optimal models.*

Hotspots form regions, not isolated points. The connected-component analysis shows that HH observations often form contiguous graph regions rather than isolated outliers. This matters in practice because regional structure is more actionable than isolated unstable predictions: it enables subgroup-level auditing, rule extraction, and targeted monitoring.

Localized multiplicity is different from standard uncertainty reporting. It is also important to distinguish the present framework from more standard uncertainty analyses. Ensemble variance, Bayesian posterior uncertainty, or repeated-training variability usually quantify uncertainty around one modeling pipeline or one stochastic training process. Rashomon multiplicity asks a different question: *how much do equally acceptable models disagree, and where does that disagreement concentrate?* A point can therefore be confidently predicted by each model individually and still be multiplicity-sensitive if acceptable models disagree systematically with one another. The hotspot analysis adds a local diagnostic layer that conventional uncertainty summaries do not provide.

5.2 Research Questions Revisited

RQ1: Where do similarly well-performing models disagree? The results show that predictive multiplicity is present at the observation level across all three benchmark datasets. The magnitude of disagreement differs by dataset: German Credit exhibits the largest aggregate multiplicity, while COMPAS and Adult show lower mean predictive variance. This confirms that near-optimal models can disagree substantially on individual observations even when they have similar validation performance.

RQ2: Does disagreement form local regions in feature space? Predictive variance is spatially structured with respect to the feature-space graph. Across all three datasets, variance-based Moran's I is positive and significant under the prediction-permutation null in nearly all runs. The spatial structure is strongest on COMPAS, where high-variance observations form clearer localized HH regions. Adult shows broader and more diffuse hotspot structure, while German Credit has high overall disagreement but weaker spatial concentration. Thus, disagreement magnitude and spatial concentration are related but distinct properties.

RQ3: How stable are these regions under changes in the analysis? The stability and robustness analyses show that hotspot structure is partly stable, but not pointwise fixed. Exact HH membership varies across seeds, and individual observations are not reliably classified as hotspots in every run. However, HH observations often form contiguous graph regions, and higher-frequency HH points remain spatially localized. The main spatial conclusions also persist under changes to the Rashomon set size K, the neighborhood size k, probability calibration, and graph construction. The appropriate interpretation is therefore regional rather than pointwise: hotspot areas recur to some extent, but their exact boundaries depend on the split, model set, and graph definition.

RQ4: Which modeling choices are associated with the instability? Model family accounts for a substantial share of observed predictive variance, especially on Adult and COMPAS, and this contribution often increases inside HH regions. Within model families, disagreement is further associated with a smaller number of grouped hyperparameter values, such as regularization strength for logistic regression, neighborhood size for kNN, and complexity or optimization parameters for tree-based and neural-network models. Predictive multiplicity is therefore not just random training variation; within the sampled candidate pool, it is systematically associated with concrete modeling choices. These results are descriptive and should not be interpreted as causal effects of individual hyperparameters.

RQ5: Can the detected regions be validated and interpreted? The synthetic experiments confirm that the pipeline can recover known ambiguous regions when ground truth is available. In both synthetic designs, LISA HH detection identifies the intended high-instability regions with high precision and substantial recall. The rule-extraction analysis further shows that real-data hotspots, especially on COMPAS, can be partially described by compact feature-space rules. Together, these results show that the framework not only detects instability, but also helps localize and describe the affected regions.

5.3 Practical Implications

In applications where retaining multiple near-optimal models is feasible, the results suggest a prospective reliability-screening workflow that differs from standard "single best model" practice:

1. train and retain a near-optimal set of models rather than only one selected winner;
2. compute pointwise multiplicity metrics and hotspot maps on held-out data;
3. treat predictions in hotspot regions differently, for example by attaching caution labels, triggering human review, or using abstention policies in high-stakes settings.

The practical contribution is therefore not merely another descriptive metric, but a way to separate *stable predictions* from *predictions that are sensitive to model choice*. In high-stakes applications, this distinction is operationally relevant. A system should not treat a prediction in a stable region and a prediction in a hotspot as equally trustworthy simply because both come with similar point estimates.

A useful way to interpret the proposed workflow is as a *reliability screening mechanism*. Predictions in stable regions may proceed through the standard pipeline. Predictions in localized hotspots should be interpreted more cautiously because the selected model represents only one of several equally accurate alternatives that may not agree on the same observation. The precise intervention will depend on the application domain, but the general principle is clear: multiplicity hotspots identify cases where model choice itself becomes decision-relevant.

This workflow should be understood as a prospective implication of the empirical findings, not as a deployment policy evaluated in this thesis. In practice, hotspot maps would need to be recomputed and validated when the data distribution, candidate model pool, or preprocessing pipeline changes. This is especially important under distribution shift, where regions that were stable on historical test data may no longer remain stable in future data.

5.4 Methodological Reflection

An auxiliary diagnostic in Appendix H compares predictive variance with distance to the ensemble decision boundary. The results suggest that boundary proximity is associated with part of the multiplicity pattern on Adult and German Credit, but not on COMPAS. This supports the broader point that spatial multiplicity need not be identical to ordinary classification ambiguity.

This distinction matters because it clarifies what kind of phenomenon the proposed pipeline is detecting. It is useful to separate three perspectives:

- **aleatoric or data ambiguity**, arising from intrinsic class overlap;
- **epistemic or model uncertainty**, arising from limited data or imperfect estimation;
- **Rashomon multiplicity**, arising because many behaviorally different models remain equally acceptable under the chosen performance criterion.

The present framework primarily targets the third category. It is therefore best understood not as a replacement for uncertainty quantification, but as a complementary diagnostic that asks whether predictions depend strongly on which acceptable model happens to be selected.
A second reflection concerns when the method is most informative. If Moran's I is near zero and hotspot regions do not emerge, then multiplicity may still be present, but it is diffuse rather than localized. In that case, global disagreement summaries may be more informative than spatial subgroup descriptions. Conversely, if an extremely large share of the dataset becomes hotspot-like, this would indicate broad instability rather than a small set of localized risk regions. The method is therefore especially valuable in the intermediate regime identified in this thesis: disagreement exists, but it is concentrated enough to support localization, explanation, and targeted intervention.

5.5 Limitations and Threats to Validity

Despite the empirical support for the proposed framework, several limitations remain.

- **Finite approximation of the Rashomon set.** The top-K candidate approach approximates a potentially much larger and more continuous set of acceptable models. The observed multiplicity structure therefore depends on the candidate families and hyperparameter spaces explored in the experiments.
- **Descriptive hyperparameter attribution.** The within-family hyperparameter importance analysis depends on the sampled candidate pool and on the grouping scheme used for hyperparameters. In particular, high-cardinality numeric hyperparameters are discretized into tertiles, so the resulting importance scores should be interpreted as descriptive group-level associations rather than causal effects of individual hyperparameter changes.
- **Dependence on the graph construction.** Although robustness checks across alternative graph constructions are encouraging, the spatial analysis still depends on how neighborhood structure is defined in transformed feature space. Different representations or manifold-aware graph constructions could reveal somewhat different local patterns.
- **Boundary instability of local hotspot labels.** Exact HH membership is unstable across seeds, even when broader hotspot regions recur. This means the method is better suited to identifying unstable *areas* than to declaring a single observation intrinsically and permanently hotspot-like.
- **External validity.** The experiments cover three benchmark datasets and synthetic validation tasks. The practical usefulness of multiplicity hotspot analysis should be further tested on larger, domain-specific, and operational datasets before drawing broad deployment conclusions.

6 Conclusion & Future Work

This thesis investigated whether predictive multiplicity—the disagreement among near-optimal models—is spatially structured in feature space. The results show that prediction instability is often not spread uniformly over the test set, but appears in localized regions. The thesis therefore treats predictive multiplicity not only as an aggregate property of a model set, but also as a local property of particular parts of the feature space.

Across the real-world benchmark datasets, predictive variance shows positive spatial autocorrelation, and high-instability observations often form local hotspot regions. The strength and interpretation of these regions differ across datasets: COMPAS shows the clearest localized structure, German Credit shows high overall disagreement with weaker spatial concentration, and Adult shows broader but more diffuse instability regions. The synthetic experiments further show that the proposed pipeline can recover known ambiguous regions under controlled conditions.

The analysis also shows that predictive multiplicity is associated with concrete modeling choices. Model family accounts for a substantial share of observed prediction variance, especially in some hotspot regions, while within-family hyperparameters add further structure. Simple rule-based descriptions indicate that selected high-multiplicity regions can be described in terms of original input features, although exact hotspot membership remains partly sensitive to the data split and graph construction.

Overall, the results suggest that model reliability should not only be assessed globally. Two models may have similar validation performance, and one deployed model may look acceptable on average, while some regions of the feature space remain highly sensitive to which near-optimal model is chosen. Spatial multiplicity analysis provides a way to identify such regions and to treat them as candidates for closer inspection.

Taken together, the findings show that predictive multiplicity is most informative when it is analyzed both globally and locally: global metrics quantify the overall extent of model disagreement, while spatial hotspot analysis identifies the regions of feature space where model choice becomes especially consequential.

6.1 Future Work

A first direction for future work is to improve the approximation of the Rashomon set. This thesis uses a fixed top-K subset of trained models because it is simple and comparable across runs. Future work could use larger candidate pools, adaptive performance thresholds, or optimization-based methods to explore the set of near-optimal models more systematically.

A second direction is to study how hotspot information could be used in practice. For example, predictions inside high-multiplicity regions could be flagged for additional review, reported with a model-sensitivity warning, or treated differently in selective prediction systems. This would require moving from retrospective analysis to deployment experiments.

A third direction is to examine stability over time. The present experiments study variation across random splits and modeling choices, but not under temporal changes in the data distribution. In applications where the population changes over time, it would be important to know whether the same high-multiplicity regions persist, move, or disappear.

Finally, the framework could be adapted beyond binary classification. In multiclass classification, regression, or ranking tasks, prediction disagreement has a different form, so the multiplicity metric and the interpretation of hotspots would need to be adjusted.

A Hyperparameter Search Spaces

Table 12 lists the hyperparameter search spaces used for training the candidate models. For each run, 50 configurations per model family are sampled from these spaces.

Table 12: Hyperparameter search spaces used for candidate model training. Parameters listed as fixed are not varied, but are included for reproducibility.

Family	**Hyperparameter**	**Values**
LogReg	`C`	20 log-spaced values from 10^{-5} to 10^{5}
	`l1_ratio`	$\{0, 0.25, 0.5, 0.75, 1.0\}$
	`penalty`	`elasticnet` (fixed)
	`solver`	`saga` (fixed)
	`max_iter`	2000 (fixed)
kNN	`n_neighbors`	$\{3, 5, 7, 10, 15, 20, 25, 30, 50, 70, 90, 100, 150\}$
	`weights`	`uniform`, `distance`
	`p`	$\{1, 2\}$
RF	`n_estimators`	$\{100, 200, 500\}$
	`max_depth`	$\{\texttt{None}, 5, 10, 20, 40\}$
	`min_samples_split`	$\{2, 5, 10\}$
	`min_samples_leaf`	$\{1, 2, 4\}$
	`max_features`	`sqrt`, `log2`, `None`
	`bootstrap`	`True` (fixed)
GBM	`n_estimators`	$\{100, 200, 500\}$
	`learning_rate`	$\{0.01, 0.05, 0.1, 0.2\}$
	`max_depth`	$\{2, 3, 4, 5\}$
	`subsample`	$\{0.6, 0.8, 1.0\}$
MLP	`hidden_layer_sizes`	(50), (100), (200), (100, 50), (200, 100)
	`alpha`	$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
	`learning_rate_init`	$\{10^{-4}, 10^{-3}, 10^{-2}\}$
	`activation`	`relu`, `tanh`
	`solver`	`adam` (fixed)
	`early_stopping`	`True` (fixed)
	`max_iter`	1000 (fixed)

B Dataset Characteristics

Table 13 summarizes the split sizes, target prevalence, and transformed feature dimensionality used in the experiments. These quantities provide context for interpreting cross-dataset comparisons, especially absolute HH counts and hotspot-specific analyses.

Table 13: Dataset characteristics after train–validation–test splitting and preprocessing. The positive rate is computed on the full dataset. The transformed dimension refers to the feature representation after imputation, standardization, and one-hot encoding.

Dataset	Total N	Train	Validation	Test	Positive rate	Transformed dim.
COMPAS	7214	4328	1443	1443	45.1%	12
German Credit	1000	600	200	200	70.0%	61
Adult	48842	29304	9769	9769	23.9%	104

C Performance-Control Meta-Model Diagnostic

As an auxiliary diagnostic, we examine whether the hyperparameter associations reported in Section 4.5 mainly reflect differences in validation performance. For each dataset–family combination, we fit a model-level meta-model predicting the disagreement contribution V_m from grouped hyperparameters and validation Brier score. This analysis is descriptive rather than causal. It is not used as the primary attribution result, because validation Brier score is a performance metric rather than a hyperparameter. Its purpose is to assess whether model-level disagreement is dominated by performance differences.

Table 14 compares the Rashomon set with the full candidate pool. In the full pool, validation Brier score is the strongest predictor in 11 of 15 dataset–family combinations, with an average importance of 0.596. This indicates that, when all trained candidates are included, validation performance differences account for a large share of model-level disagreement. Within the Rashomon set, this effect is weaker: validation Brier score is rank 1 in 7 of 15 combinations, and its mean importance decreases to 0.290. At the same time, the strongest non-performance hyperparameter has a higher mean importance in the Rashomon set (0.498) than in the full pool (0.285).

These results suggest that the Rashomon restriction reduces, but does not eliminate, the role of performance differences. Disagreement inside the near-optimal set is therefore not simply a byproduct of poorly performing models. However, validation Brier score remains important in several dataset–family combinations, especially for GBM, MLP, and parts of kNN. The meta-model analysis is therefore interpreted as a performance-control diagnostic rather than as primary evidence for hyperparameter importance.

Table 14: Meta-model diagnostic summary comparing the Rashomon set with the full candidate pool. Values are aggregated over 15 dataset–family combinations. “Brier rank-1” reports in how many combinations validation Brier score is the most important predictor of V_m.

Pool	Groups	Brier rank-1	Mean Brier importance	Mean top HP importance
Rashomon	15	7	0.290	0.498
Full pool	15	11	0.596	0.285

D Additional Rule-Extraction Results

Table 15: Representative global HH rules on COMPAS.

Seed	Rule	Support	Purity	Recall	Lift
2	priors_count > 20.5	28	1.000	0.230	11.83
6	priors_count > 17.5 AND age > 34.5	29	1.000	0.199	9.88
7	age > 25.5 AND priors_count > 18.5	31	0.968	0.171	7.98
5	priors_count > 19.5 AND age > 36.5	27	0.926	0.352	18.82
8	age > 40.5 AND priors_count > 14.5	33	0.879	0.147	6.44

Table 16: Feature frequency across global HH rules on COMPAS.

Feature	Rules	Seeds	Mean purity	Mean lift
age	58	10	0.321	4.577
priors_count	54	10	0.332	4.697
sex_Female	6	3	0.168	1.472
race_Caucasian	5	2	0.433	2.640
race_Other	4	2	0.134	2.671
c_charge_degree_F	2	1	0.655	7.751
c_charge_degree_M	2	1	0.080	0.657
sex_Male	1	1	0.154	0.819

E Synthetic FDR Sensitivity

To assess whether the synthetic validation results depend on the multiple-testing threshold used for LISA, we vary the false-discovery-rate level over $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$.

Single-island design. In the single-island setting, the results are stable across the tested thresholds. As α increases, the number of HH points rises moderately from 88 to 109, while the Jaccard overlap with the ground-truth island increases from 0.72 to 0.79. This reflects the expected precision–recall trade-off: precision decreases only slightly, from about 0.99 at $\alpha = 0.01$ to about 0.93 at $\alpha = 0.20$, whereas recall increases from 0.73 to 0.84. Across all tested α values, the detected HH region remains a single connected component, matching the geometry of the true island.

Three-islands design. In the three-islands setting, the conclusions are also stable across thresholds. At the strictest threshold $\alpha = 0.01$, the method detects 265 HH points with precision 0.9623, recall 0.8673, and Jaccard overlap 0.8388. Increasing the threshold to $\alpha = 0.05$ raises the HH count slightly to 268 and improves recall to 0.8776, with Jaccard overlap increasing to 0.8487 while precision remains essentially unchanged. The results are identical at $\alpha = 0.10$ and change only marginally at $\alpha = 0.20$, where the HH count increases to 269, precision decreases slightly to 0.9591, recall remains 0.8776, and Jaccard overlap decreases slightly to 0.8459. Across all tested thresholds, the number of connected components remains four, reflecting the three island hotspots plus one additional outlier-driven hotspot component.

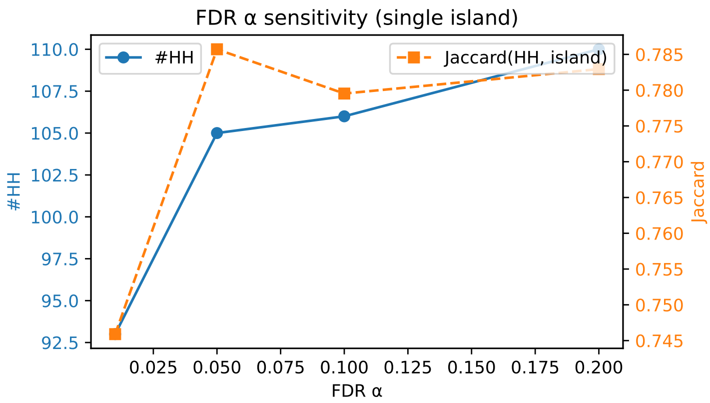


Figure 14: FDR sensitivity analysis for the single-island design. HH count and Jaccard overlap are shown as a function of the LISA false-discovery-rate threshold α.

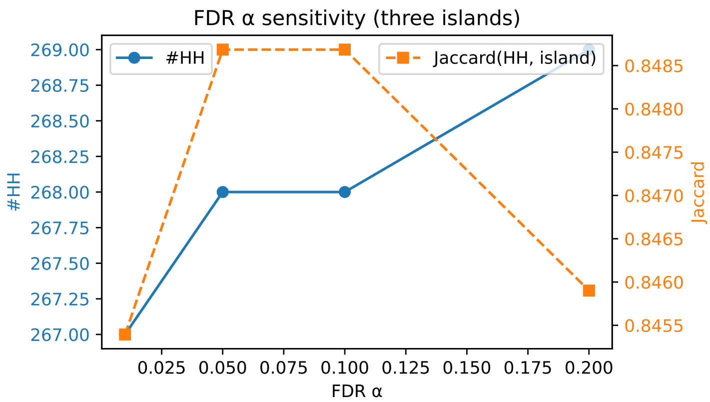
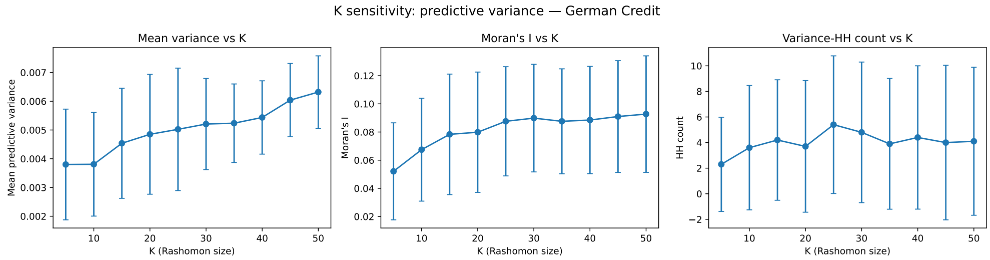


Figure 15: FDR sensitivity analysis for the three-islands design. HH count and Jaccard overlap are shown as a function of the LISA false-discovery-rate threshold α.

F Additional Sensitivity Curves

The main text reports the sensitivity curves for COMPAS, which is used as the running example for the detailed robustness analysis. For completeness, Figures 16 and 17 show the corresponding variance-based sensitivity curves for German Credit and Adult.

(a) German Credit



(b) Adult

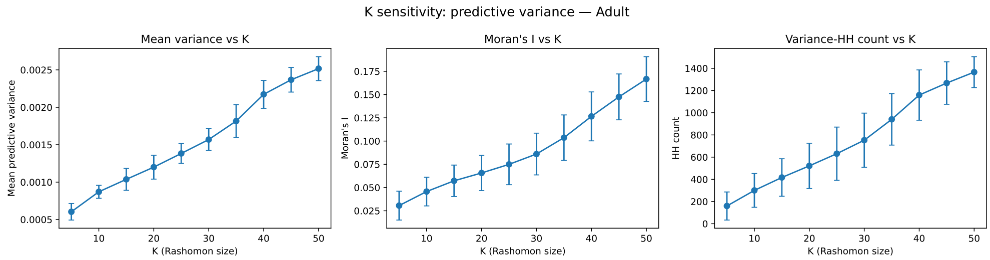


Figure 16: Sensitivity to Rashomon set size K for German Credit and Adult (mean $\pm$ standard deviation over runs). Panels show mean predictive variance, Moran's I, and variance-based HH count.

G Decision-Level Multiplicity: Conflict Ratio

The main analysis in this thesis uses predictive variance as the primary multiplicity metric, because it preserves probability-space disagreement between near-optimal models. As a supplementary diagnostic, we also compute a decision-level conflict ratio. This metric captures whether models in the Rashomon approximation fall on different sides of the classification threshold.

Let $\tau = 0.5$ be the decision threshold. For each observation i, define

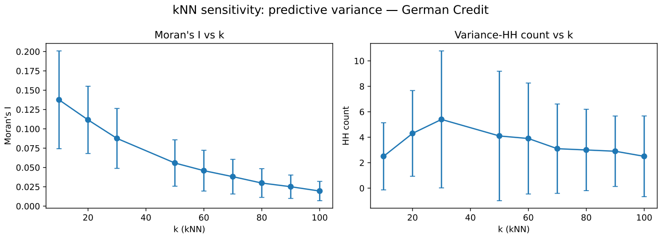
$$q_i = \frac{1}{K} \sum_{m=1}^{K} \mathbf{1}[\hat{p}_{mi} \geq \tau]$$

as the fraction of models in the top-K Rashomon approximation that predict the positive class. The conflict ratio is then

$$c_i = \min(q_i, 1 - q_i), \qquad c_i \in [0, 0.5].$$

A value of $c_i = 0$ indicates unanimous agreement in the thresholded class label, whereas $c_i = 0.5$ indicates a maximally split model set. In contrast to predictive variance, which

(a) German Credit



(b) Adult

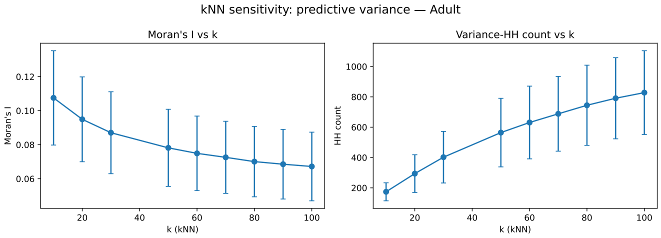


Figure 17: Sensitivity to kNN neighborhood size k for German Credit and Adult (mean $\pm$ standard deviation over runs). Panels show Moran's I and variance-based HH count.

captures disagreement in predicted probabilities, the conflict ratio only captures disagreement that affects the final class decision.

G.1 Dataset-Level Conflict Metrics

Table 17 summarizes conflict-related metrics across datasets. The results broadly mirror the variance-based analysis, but also show that disagreement magnitude and spatial concentration are not identical. German Credit has the largest overall conflict, but its conflict-based Moran's I is comparatively small. COMPAS has lower overall conflict prevalence, but stronger spatial concentration. Adult has the lowest overall conflict magnitude, while still showing positive conflict-based spatial clustering.

Table 17: Conflict metrics by dataset (mean $\pm$ std over 10 runs, global top-K ($K = 25$), dataset-specific k).

Dataset	Mean c_i	$c_i > 0$ (%)	$c_i \geq 0.25$ (%)	Conflict Moran's I	Conflict HH
COMPAS	0.035 ± 0.007	17.9 ± 2.9	5.9 ± 1.6	0.17 ± 0.03	122.0 ± 23.4
German Credit	0.059 ± 0.022	30.1 ± 8.5	9.8 ± 4.6	0.05 ± 0.03	1.3 ± 2.4
Adult	0.017 ± 0.001	9.5 ± 0.5	2.7 ± 0.2	0.12 ± 0.01	627.1 ± 59.9

G.2 Overlap with Variance-Based Hotspots

The overlap between variance-based and conflict-based HH hotspots is limited across all datasets. On COMPAS, the Jaccard similarity is 0.078 ± 0.057, indicating that the two metrics identify largely different observations as hotspots. Adult shows somewhat higher but still limited overlap (0.184 ± 0.052). On German Credit, the mean overlap is 0.118 ± 0.312, but this estimate is unstable because both hotspot sets are often very small. This confirms that predictive variance and conflict ratio capture related but distinct forms of multiplicity. Predictive variance measures probability-level instability, while the conflict ratio isolates threshold-sensitive decision disagreement.

G.3 Quadrant Analysis

To compare both metrics more directly, observations are partitioned into four quadrants using the 90th percentile of predictive variance and conflict ratio within each run. This yields four regimes: jointly high variance and conflict, high variance only, high conflict only, and jointly low disagreement.

Table 18: Quadrant breakdown for COMPAS (mean ± std over 10 runs, global top-K ($K = 25$)). Thresholds are the 90th percentile of variance and conflict per run.

Quadrant	Number of test observations	Share of test observations	Predictive variance v_i	Conflict ratio c_i
A (high var, high conflict)	33.9 ± 8.2	2.3% ± 0.6%	0.0068 ± 0.0017	0.290 ± 0.057
B (high var, low conflict)	111.6 ± 8.1	7.7% ± 0.6%	0.0043 ± 0.0012	0.010 ± 0.007
C (low var, high conflict)	125.3 ± 10.0	8.7% ± 0.7%	0.0011 ± 0.0003	0.278 ± 0.051
D (low var, low conflict)	1172.2 ± 9.9	81.2% ± 0.7%	0.0009 ± 0.0002	0.005 ± 0.003

Most observations fall into the stable regime D. Among the remaining observations, quadrants B and C are both more common than quadrant A. This shows that high disagreement often appears in only one of the two metrics: either as broad probability variation without label changes, or as threshold-sensitive label instability without especially large probability spread.

G.4 Conflict-Based Null Testing

We also repeat the prediction-permutation null experiment using the conflict ratio c_i instead of predictive variance. The results are shown in Table 19. Conflict-based Moran's I is significant in all runs for COMPAS and Adult, and in most runs for German Credit.

Table 19: Conflict-based prediction-permutation null summary by dataset. A run is counted as significant when the empirical permutation p-value for conflict Moran's I is below 0.05. The null uses $R = 100$ prediction-matrix permutations per run. Values are computed over 10 runs.

Dataset	Runs	Significant runs	Conflict Moran's I
Adult	10	100%	0.118 ± 0.013
COMPAS	10	100%	0.167 ± 0.033
German Credit	10	90%	0.049 ± 0.029

Thus, a spatial signal is also visible for thresholded decision disagreement. However, because the main thesis focuses on probability-space disagreement, the conflict-ratio analysis is treated as supplementary rather than as part of the main empirical argument.

H Decision-Boundary Proximity

As an auxiliary diagnostic, we test whether variance-based hotspots mainly occur near the ensemble decision boundary. For each test observation, we compute the margin

$$m_i = |\bar{p}_i - 0.5|,$$

where $\bar{p}_i$ is the mean predicted probability across the selected Rashomon models. Lower values of m_i indicate observations closer to the decision boundary. We summarize this relationship using Pearson and Spearman correlations between pointwise predictive variance and margin. In addition, we compare the margin distributions of HH and non-HH observations using a one-sided Mann–Whitney U test.

Table 20 shows a clear cross-dataset contrast. On Adult and German Credit, predictive variance and margin are negatively correlated, meaning that high-variance observations tend to lie closer to the decision boundary. This pattern is especially pronounced in rank terms: the mean Spearman correlation is -0.89 on Adult and -0.70 on German Credit. The HH comparison points in the same direction: HH observations have smaller average margins than non-HH observations on both datasets.

COMPAS behaves differently. The mean Pearson correlation is -0.04 ± 0.09 and the mean Spearman correlation is -0.07 ± 0.18, both close to zero. HH observations also do not have smaller margins than non-HH observations; their mean margin is slightly larger (0.19 vs. 0.18). Thus, COMPAS hotspots are not well explained by ordinary decision-boundary proximity.

Table 20: Variance–margin relationship by dataset (mean $\pm$ std across runs). Margin is defined as $m_i = |\bar{p}_i - 0.5|$, so lower values indicate observations closer to the decision boundary. HH and non-HH columns report the average margin inside and outside variance-based HH hotspots.

Dataset	Pearson r	Spearman ρ	Margin in HH	Margin outside HH
Adult	-0.33 $\pm$ 0.04	-0.89 $\pm$ 0.02	0.20 $\pm$ 0.01	0.38 $\pm$ 0.01
COMPAS	-0.04 $\pm$ 0.09	-0.07 $\pm$ 0.18	0.19 $\pm$ 0.04	0.18 $\pm$ 0.00
German Credit	-0.56 $\pm$ 0.13	-0.70 $\pm$ 0.10	0.13 $\pm$ 0.05	0.26 $\pm$ 0.02

The Mann–Whitney U tests support the same interpretation. On Adult, the alternative hypothesis that HH observations have lower margins than non-HH observations is significant in all runs (mean $p \approx 3.5 \times 10^{-22}$). On German Credit, the effect is weaker but still present, with significance in 57% of runs. On COMPAS, the test is significant in only 30% of runs and the mean p-value is 0.59, providing no evidence that HH observations systematically lie closer to the decision boundary.

Overall, the diagnostic suggests that predictive multiplicity has different mechanisms across datasets. On Adult and German Credit, high-variance regions are more closely linked to boundary ambiguity. On COMPAS, predictive variance is largely decoupled from decision-boundary proximity, suggesting that its hotspot structure is driven more by structural disagreement among near-optimal models than by closeness to the classification threshold.

I Fairness and Subgroup Exposure on COMPAS

As an exploratory auxiliary diagnostic, we examine whether variance-based HH regions on COMPAS are unevenly distributed across sensitive groups defined by race and sex. For each group, we compute the exposure rate as the fraction of that group's test observations classified as HH. Exposure rates are computed separately for each seed and then aggregated across seeds.

Table 21 shows subgroup-level differences in HH exposure. African-American test points exhibit a higher mean HH rate than Caucasian and Hispanic test points. However, some groups have small sample sizes and correspondingly high variability, so these raw differences should be interpreted cautiously.

Table 21: COMPAS: HH rates by subgroup (mean ± std across 10 seeds). Larger groups are more reliable.

Attribute	Subgroup	Test observations	HH rate	Predictive variance
Race	African-American	729	11.4% ± 7.5%	0.0013 ± 0.0004
	Caucasian	504	5.6% ± 2.8%	0.0012 ± 0.0002
	Hispanic	128	3.5% ± 3.5%	0.0012 ± 0.0003
	Other	71	14.3% ± 13.0%	0.0019 ± 0.0005
Sex	Female	279	8.5% ± 8.7%	0.0014 ± 0.0003
	Male	1164	9.0% ± 4.6%	0.0013 ± 0.0003

To test whether subgroup exposure differences are larger than expected by chance, we use a stratified permutation test. Group labels are shuffled within each run, run-level exposure rates are recomputed, and the aggregated inter-group disparity, defined as the max–min HH rate across eligible groups, is compared with its permutation null distribution. To reduce instability from very small groups, we apply per-run subgroup minimum-size filtering and require minimum cross-run support before interpreting a subgroup.

For race, the observed disparity is about 0.094, compared with a permutation-null mean of 0.012 and a 95% null interval of approximately $[0.003, 0.026]$. The resulting p-value is $p \approx 0.0005$, indicating that race-based HH exposure differences are larger than expected under this null. For sex, the observed disparity is essentially zero (0.00015), with $p \approx 0.98$, providing no evidence of sex-based HH exposure disparity.

This suggests that multiplicity exposure on COMPAS may be unevenly distributed across racial groups. However, the result should be interpreted as an exploratory subgroup auditing signal rather than as a causal fairness conclusion. The analysis is observational, and some race categories contain relatively few observations.

J Electronic appendix

The electronic appendix contains the code, generated intermediate results, tables, and figures used to reproduce the empirical analysis and thesis assets. The code repository is available at:

`https://github.com/DejvisT/rashomon-multiplicity`

The main repository is organized around the following entry points:

- `run_training_pipeline.py`: trains candidate models, ranks them by validation Brier score, constructs the finite top-K Rashomon approximations, and saves prediction matrices and run-level summaries.
- `run_training_pipeline_fixed_test.py`: repeats the training pipeline with a fixed test set for the cross-seed hotspot stability analysis.
- `analysis/`: contains the main analysis modules, including multiplicity metrics, spatial analysis, calibration robustness, family and hyperparameter decomposition, and rule extraction.
- `notebooks/`: contains exploratory and analysis notebooks used during development. Final thesis-facing tables and figures are exported through the script listed below.
- `scripts/export_thesis_assets.py`: exports the LaTeX tables and PDF figures included in the thesis from the stored experiment outputs.
- `overleaf_bundle/presentation_assets/`: contains the generated tables and figures included by the LaTeX thesis.
- `requirements.txt`: records the Python package dependencies used for the experiments.

The experiments use Python with `scikit-learn` for model training and `libpysal/esda` from PySAL for graph-based spatial statistics. Random seeds are fixed throughout the pipeline to support reproducibility.

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Use of AI Tools

I hereby confirm that I have written all parts of this thesis myself and that I take full responsibility for its content, including the methodological choices, experimental results, interpretations, and submitted program code. During the preparation of this thesis, I used generative AI tools as supporting aids.
I used OpenAI ChatGPT for linguistic and structural support, including suggestions for improving sentence clarity, shortening or reorganizing paragraphs, refining section transitions, and checking whether the argumentation of individual sections was coherent. Suggested formulations were manually reviewed, edited, and adapted by me.
I also used ChatGPT as a programming and debugging assistant during the development of the experimental pipeline. In particular, it supported the explanation of error messages, the refactoring of Python scripts and notebooks, the design of plotting and table-export routines, and the review of code. The final code was executed, tested, and revised by me, and all reported numerical results are based on my own experimental runs.
In addition, I used Cursor IDE with AI assistance for code editing, refactoring, and project organization. This included support for restructuring scripts, improving reproducibility, reducing duplicated code, and generating or revising helper functions. All AI-generated code suggestions were reviewed, modified where necessary, and integrated only after I had checked that they were consistent with the intended experimental design.
For literature-related work, I used AI tools only as an orientation aid to check whether there were additional related papers that I might have missed during my own literature search. Any suggested sources were subsequently checked against the original publications before being cited or discussed in the thesis. The selection, interpretation, and integration of the literature were carried out by me.

Declaration of authorship

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